

FINRCA: Anomaly-Propagation-Aware Lineage Graph for Automated Root Cause Analysis in Financial Risk Pipelines

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Abstract (영문 초록)

Modern financial institutions are required under BCBS 239 to maintain complete, accurate, and traceable data lineage across their risk reporting pipelines. Yet as of 2025, fewer than 10% of globally systemically important banks fully comply, with data lineage cited as the most persistent technical challenge. When anomalies occur in multi-stage financial risk pipelines—spanning raw market data ingestion, ETL transformations, feature engineering, and regulatory reporting—identifying the root cause manually requires hours to days of expert investigation.

This thesis proposes **FINRCA**, a system for automated root cause analysis in financial risk pipelines. FINRCA automatically constructs a column-level **Financial Risk Lineage Graph (FRLG)** from a pipeline definition, capturing a five-layer node hierarchy (Source, ETL, Feature, Risk, Report) and four transform-typed edges (DirectMap, Filter, Calculate, Aggregate). An **Anomaly-Propagation-Aware Root Cause Analysis (APA-RCA)** algorithm runs on the graph, extending Random Walk with Restart with hop-distance-adaptive weighting, transform-type-based edge weights, and ancestor-constrained ranking. A five-signal anomaly detector—combining rolling z-score, short-long variance ratio, segment variance, peer deviation, and causal excess isolation—provides the node-level anomaly scores. A complementary **Schema-Constraint RCA** module detects aggregation schema violations with 100% Top-1 accuracy and zero false positives.

We construct the **Synthetic Financial Risk Pipeline (SynFRP)** benchmark: 630 controlled trials across seven anomaly types and three pipeline scales, providing ground-truth root cause labels. APA-RCA v2 achieves 77.9% overall Top-5 accuracy, outperforming the anomaly-score-only baseline by 7.7 percentage points. An **Anomaly Transfer Function (ATF)** framework provides a formal model of anomaly propagation through pipeline transforms, deriving principled algorithm parameters and a distribution-free temporal discriminant that achieves perfect A3/A5 separation on both synthetic and real data.

To validate real-market robustness, we design the **Real-Source Hybrid Benchmark (RSHB)**: 12 instruments (US equities, Korean Exchange equities, FX, Treasury rates) over 756 trading days spanning three market regimes, achieving 82.4% Top-5 accuracy on real market data.

Finally, we address the challenge of incorporating machine learning without labeled anomaly data—a structural constraint of the financial domain. A **VAE/Conditional VAE (C-VAE)/Hybrid TSD-CVAE anomaly scorer**, trained unsupervised on approximately 898,000 windows from anomaly-free pipelines, replaces the z-score detector. A **GNN/ResGCN re-ranker**, trained on synthetic injection locations and applied to real data without fine-tuning, achieves 54.8% Top-1 accuracy on RSHB—a +21.9 percentage point improvement over APA-RCA v2 alone. An architecturally enhanced v2 pipeline (C-VAE + ResGCN), which encodes ATF domain theory directly into the model structure, further improves Top-3 accuracy to 74.8% (+9.1 pp) and MRR to 0.626. A further architectural refinement, v3 (Hybrid TSD-CVAE + ResGCN), introduces type-specific decoder sub-networks while preserving conditioning, achieving 80.0% Top-3 (+5.2 pp over v2), 83.3% Top-5, and the highest MRR of 0.639. An ablation study comparing three unsupervised scorers (VAE, IsolationForest, OneClass-SVM) confirms that VAE reconstruction error is the only scorer that provides transferable GNN features across the synthetic-to-real distribution shift: IsolationForest achieves 0.5% and OneClass-SVM achieves 0.0% Top-1, while VAE achieves 54.8%.

Keywords: financial risk pipeline, data lineage, root cause analysis, anomaly detection, variational autoencoder, graph neural network, BCBS 239, random walk with restart

국문 초록

BCBS 239 규정에 따라 글로벌 시스템적 중요 은행(G-SIB)은 리스크 보고 파이프라인 전반에 걸쳐 완전하고 정확하며 추적 가능한 데이터 계보(lineage)를 유지해야 한다. 그러나 2025년 현재, 전체 G-SIB 중 완전 준수 기관은 10% 미만이며, 데이터 계보 관리는 가장 지속적인 기술적 과제로 지목되고 있다. 원시 시장 데이터 수집, ETL 변환, 피처 엔지니어링, 규제 보고에 이르는 다단계 금융 리스크 파이프라인에서 이상 현상이 발생했을 때, 근본 원인을 수동으로 식별하는 데는 수 시간에서 수 일이 소요된다.

본 논문은 금융 리스크 파이프라인에서의 자동 근본 원인 분석 시스템인 **FINRCA**를 제안한다. FINRCA는 파이프라인 정의로부터 컬럼 수준의 **금융 리스크 계보 그래프(FRLG)**를 자동 구성한다. FRLG는 5계층 노드 구조(Source, ETL, Feature, Risk, Report)와 4가지 변환 유형 에지(DirectMap, Filter, Calculate, Aggregate)를 포착한다. 그래프 위에서 **이상 전파 인식 근본 원인 분석(APA-RCA)** 알고리즘이 실행되며, 홉 거리 적응형 가중치, 변환 유형 기반 에지 가중치, 조상 제약 순위화를 통해 재시작 확률을 갖는 랜덤 워크(RWR)를 확장한다. 롤링 z-점수, 단기-장기 분산 비율, 구간 분산, 피어 편차, 인과 초과 격리의 5가지 신호를 결합한 이상 감지기가 노드 수준의 이상 점수를 산출한다. 보완적인 **스키마 제약 RCA** 모듈은 집계 스키마 위반을 Top-1 정확도 100%, 오탐률 0%로 탐지한다.

평가를 위해 **SynFRP** 벤치마크를 구축하였다: 7가지 이상 유형과 3가지 파이프라인 규모에 걸쳐 630회의 통제된 실험으로 구성되며 정답 레이블을 제공한다. APA-RCA v2는 전체 Top-5 정확도 77.9%를 달성하여 이상 점수 단독 기준선을 7.7 퍼센트포인트 상회한다. **이상 전달 함수(ATF)** 프레임워크는 파이프라인 변환을 통한 이상 전파의 수학적 모델을 제공하고, 알고리즘 파라미터에 대한 원리적 근거 및 합성·실제 데이터 모두에서 완벽한 분리 성능을 달성하는 분포-무관 시간적 판별자를 도출한다.

실제 시장 견고성 검증을 위해 **실소스 하이브리드 벤치마크(RSHB)**를 설계하였다: 세 가지 시장 국면에 걸친 756 거래일 동안 12개 금융 상품(미국 주식, 한국거래소 주식, 외환, 국채 금리)을 포함하며, Top-5 정확도 82.4%를 실제 시장 데이터에서 달성한다.

마지막으로, 이상 레이블 데이터 없이 머신러닝을 통합하는 방법론을 제안한다. 이상이 없는 파이프라인의 약 89만 8천 개 윈도우로 비지도 학습된 **VAE/조건부 VAE(C-VAE) 이상 스코어러**가 z-점수 탐지기를 대체한다. 합성 주입 위치를 학습 레이블로 사용하고 파인튜닝 없이 실제 데이터에 적용되는 **GNN/ResGCN 재순위화기**는 RSHB에서 Top-1 정확도 54.8%를 달성하며, 이는 APA-RCA v2 단독 대비 +21.9 퍼센트포인트 향상이다. ATF 도메인 이론을 모델 구조에 직접 반영한 아키텍처 강화 v2 파이프라인(C-VAE + ResGCN)은 Top-3 정확도를 74.8%(v1 대비 +9.1pp)로, MRR을 0.626으로 추가 향상시킨다. 추가적인 아키텍처 개선인 v3(Hybrid TSD-CVAE + ResGCN)는 변환 유형별 전용 디코더를 도입하면서 conditioning을 유지하여, Top-3 정확도 80.0%(v2 대비 +5.2pp), Top-5 83.3%, 최고 MRR 0.639를 달성한다. VAE, IsolationForest, OneClass-SVM을 비교하는 ablation 연구는 VAE 재구성 오차만이 합성·실제 분포 이동을 초월하는 전이 가능한 GNN 피처를 제공함을 실험으로 확인한다.

핵심어: 금융 리스크 파이프라인, 데이터 계보, 근본 원인 분석, 이상 탐지, 변분 오토인코더, 그래프 신경망, BCBS 239, 재시작 확률을 갖는 랜덤 워크

Chapter 1. Introduction

1.1 Research Background

1.1.1 The BCBS 239 Compliance Gap

In January 2013, the Basel Committee on Banking Supervision published BCBS 239, formally titled Principles for effective risk data aggregation and risk reporting. The regulation mandates that globally systemically important banks (G-SIBs) ensure the accuracy, completeness, and timeliness of their risk data, and demonstrate end-to-end traceability from raw source systems to regulatory reports. The deadline for initial compliance was January 2016.

More than a decade later, the gap between regulation and practice remains striking. The Bank for International Settlements' 2025 implementation review reports that among 31 G-SIBs, only two institutions are in full compliance, and no single principle has achieved universal compliance across the peer group. The European Central Bank's 2024 Risk Data Aggregation and Risk Reporting guide explicitly names attribute-level data lineage as one

of seven supervisory priority areas requiring sustained attention. Industry surveys confirm that structural data governance deficiencies persist across the majority of G-SIBs, with data lineage cited as the single most persistent technical challenge.

The practical difficulty is real. Financial risk pipelines are multi-stage systems in which raw market data—equity prices, interest rates, foreign exchange rates, credit spreads—undergoes dozens of sequential transformations before reaching regulatory reports. These stages include data cleaning and normalization, currency conversion, feature engineering, risk metric calculation (VaR, Expected Shortfall, sensitivities), and compliance reporting. Each stage introduces potential failure points. When a reported Value-at-Risk figure is incorrect, identifying which upstream transformation introduced the error requires traversing a complex dependency graph, a task that today remains largely manual and may take hours to days of investigation by skilled quantitative analysts.

1.1.2 Heightened Stakes Under FRTB 2025

The Fundamental Review of the Trading Book (FRTB), mandatory for G-SIBs from January 2025, substantially increases both the computational complexity and the auditability requirements of risk pipelines. Two FRTB provisions directly amplify the need for automated data lineage root cause analysis.

First, the **P&L Attribution Test (PLAT)** requires firms to demonstrate monthly that their Internal Model Approach risk-theoretical P&L closely tracks the hypothetical P&L computed from front-office pricing. Failure causes automatic fallback to the Standardized Approach, which typically carries 40-60% higher capital requirements. A PLAT failure is, at its root, a data lineage discrepancy: some upstream transformation produced risk figures inconsistent with the front-office source of truth. Identifying which pipeline node is responsible is precisely the problem this research addresses.

Second, FRTB's **Sensitivity-Based Approach** mandates risk factor sensitivity calculations at the individual instrument level, adding computation layers that were previously aggregated. This increases node counts in existing VaR pipelines proportionally, making manual root cause analysis even more burdensome.

These regulatory developments create a clear and urgent need for automated, interpretable root cause analysis tooling that can operate at financial pipeline scale and latency requirements.

1.1.3 The Fundamental Data Scarcity Problem

A distinctive challenge in applying machine learning to financial pipeline root cause analysis is the near-total absence of labeled anomaly data. Financial institutions experience pipeline failures infrequently—by design, given their operational criticality—and when failures occur, they are rarely labeled, classified, and archived in a machine-readable format suitable for supervised learning.

This is not merely a practical inconvenience. The Basel Committee's own surveys indicate that the majority of G-SIBs lack systematic incident tracking for data quality events in their risk pipelines. Commercial financial data providers such as Bloomberg Terminal (approximately USD 25,000 per terminal per year, institutional contracts only) and Refinitiv Eikon provide raw market time series but carry no anomaly annotations. Korean Exchange (KRX) data contracts are restricted to institutional counterparties and similarly provide no labeled failure events.

The consequence is that purely supervised machine learning approaches—classifiers trained on historical labeled incidents—are structurally infeasible for this domain. Any viable machine learning component must operate without relying on real labeled anomaly data.

1.2 Research Motivation

The author's direct experience at KB Kookmin Bank's StarBanking Operations Team provided concrete motivation for this research. When discrepancies arose in reported risk metrics, operational teams spent hours to days tracing data flows across disparate systems, relying on institutional knowledge and ad hoc SQL queries rather than systematic tooling. Root cause identification was a manual, expertise-dependent process with no reproducible methodology.

This experience motivates the central research question:

Can the root cause of a data quality anomaly in a financial risk reporting pipeline be identified automatically, rapidly, and accurately using a data lineage graph—without requiring labeled anomaly data?

1.3 Research Objectives and Scope

This thesis pursues four objectives:

Objective 1 — Formal Representation. Define a column-level lineage graph schema specifically designed for financial risk computation pipelines, capturing the five-layer node hierarchy and four-type edge taxonomy that characterize VaR/ES computation systems.

Objective 2 — Automated RCA Algorithm. Design and implement a root cause analysis algorithm that exploits both graph structure and node-level anomaly signals to rank candidate root causes, with emphasis on interpretability and sub-second latency.

Objective 3 — Reproducible Evaluation Infrastructure. Construct a reproducible benchmark with controlled anomaly injection across a taxonomy of seven financial anomaly types, and validate against real market data from multiple equity markets.

Objective 4 — ML Integration Without Label Dependency. Investigate how machine learning components can augment the RCA pipeline without requiring labeled anomaly data, and empirically characterize which unsupervised methods can generalize across synthetic-to-real distribution shifts.

Scope

The system targets **financial risk data pipelines** in the sense of BCBS 239: multi-stage computational systems that transform raw market data into regulatory risk reports. The anomaly taxonomy covers seven types spanning source data corruption, ETL logic errors, feature calculation bugs, aggregation errors, downstream masking, multi-source interference, and coincidental anomalies. The evaluation covers pipelines of 45-100+ nodes. Real-time streaming pipelines and high-frequency trading systems are outside scope.

1.4 Contributions

This thesis makes six contributions:

C1 — Financial Risk Lineage Graph (FRLG). A column-level lineage graph schema for financial risk pipelines, capturing five node types (Source, ETL, Feature, Risk, Report) and four edge types (DirectMap, Filter, Calculate, Aggregate). This is, to the author’s knowledge, the first lineage schema specifically designed for financial risk computation as opposed to general-purpose metadata catalogs.

C2 — Hop-Distance-Adaptive APA-RCA. An Anomaly-Propagation-Aware Root Cause Analysis algorithm that extends Random Walk with Restart with a novel hop-distance-adaptive weighting scheme. Unlike all prior graph-walk RCA methods (MicroRCA, CloudRanger, CORAL, MULAN, REASON), which use fixed transition weights, APA-RCA v2 adapts restart probability based on hop distance from the observed anomaly target. A companion five-signal anomaly detector combines causal excess isolation, peer deviation scoring, short-long variance ratio, rolling z-score, and segment variance signals.

C3 — Schema-Constraint RCA and ML Enhancement Pipeline. A lightweight $O(|V|)$ structural integrity check on the FRLG that detects aggregation schema violations (A4 anomaly type) with 100% Top-1 accuracy and zero false positives on non-A4 types. Additionally, a three-version ML enhancement pipeline: v1 (VAE + GNN), v2 (C-VAE + ResGCN), and v3 (Hybrid TSD-CVAE + ResGCN), progressively encoding ATF domain theory into the model architecture through conditioning design (v2) and type-specific decoder routing (v3).

C4 — SynFRP Benchmark. A configurable synthetic financial risk pipeline benchmark with GBM/Vasicek/Random-Walk data generation and a controlled seven-type anomaly injection protocol, providing ground truth for 630 experimental runs across three pipeline scales.

C5 — Anomaly Transfer Function (ATF) Framework. A formal model of anomaly propagation through financial pipeline transforms, yielding closed-form transfer coefficients per transform type. The ATF provides principled values for APA-RCA’s parameters and derives a distribution-free temporal discriminant that achieves perfect separation of A3 (feature calculation bug) from A5 (downstream masking) on both synthetic and real data.

C6 — Real-Source Hybrid Benchmark (RSHB) and ML Validation. An expanded real-market evaluation benchmark covering 12 instruments across US equities, Korean Exchange equities, foreign exchange, and Treasury rates over 756 trading days spanning three market regimes. The RSHB serves as the cross-domain transfer evaluation dataset for a VAE + GNN ML enhancement pipeline, demonstrating +21.9 percentage points Top-1 improvement without fine-tuning on real data. The v3 pipeline (Hybrid TSD-CVAE + ResGCN) achieves the highest Top-3 of 80.0% (+14.3 pp over v1) and MRR of 0.639, while also recovering A4 accuracy from 53.3% (v2) to 70.0% through type-specific decoder separation.

1.5 Thesis Organization

The remainder of this thesis is structured as follows.

Chapter 2 surveys related work in four areas: data lineage and provenance systems, root cause analysis methods, graph neural networks applied to RCA, and unsupervised anomaly detection methods.

Chapter 3 defines the system design and problem formulation: the financial risk pipeline model, the FRLG schema, the multi-signal anomaly detector, the Anomaly Transfer Function framework, and the evaluation benchmarks (SynFRP and RSHB).

Chapter 4 presents the APA-RCA algorithm in detail, including the hop-distance-adaptive RWR formulation, the Schema-Constraint RCA module, and the OpenLineage compatibility layer.

Chapter 5 describes the ML-enhanced pipeline: the VAE/C-VAE anomaly scorer, the GNN/ResGCN re-ranker, the ATF-aligned feature design, the cross-domain transfer protocol, and the ablation study comparing VAE against alternative unsupervised scorers (IsolationForest, OneClass-SVM).

Chapter 6 presents experimental results on the SynFRP benchmark and the RSHB, including per-type breakdown, scalability analysis, and architectural enhancement analysis.

Chapter 7 concludes with a summary of findings, limitations, and directions for future research.

Chapter 2. Related Work

2.1 Data Lineage and Provenance Systems

2.1.1 Foundational Provenance Research

The concept of data provenance—tracking the origin and transformation history of data—has been studied in the database and scientific workflow communities since the early 2000s. Cheney et al. provide a foundational survey distinguishing three classes of provenance: why-provenance (which input tuples contributed to an output), where-provenance (which input cells were copied), and how-provenance (the transformation derivation tree). Davidson and Freire survey provenance in scientific workflows, establishing the importance of lineage for reproducibility and error diagnosis.

These foundational works establish that data lineage graphs—directed acyclic graphs in which nodes represent data artifacts or transformations and edges represent derivation relationships—are the canonical data structure for provenance tracking. This thesis adopts this representation as the basis for the FRLG.

2.1.2 Industrial Lineage Platforms

Industrial lineage tools have matured substantially in the past decade. Apache Atlas provides a metadata governance framework for Hadoop ecosystems, with lineage tracking at the dataset level. OpenLineage defines an open standard for capturing lineage events emitted by data pipelines (Airflow, Spark, dbt), with a schema based on Job (transformation) and Dataset (data artifact) entities. Atlan and Solidatus offer commercial data catalog products with lineage visualization capabilities.

These platforms share a common limitation relative to the goals of this thesis: **none provides root cause analysis capability**. They record what lineage exists but offer no automated method for identifying which node caused an observed anomaly. Furthermore, none has been evaluated with published RCA accuracy metrics. This thesis bridges the gap between lineage recording (what existing tools do) and lineage exploitation for diagnosis (what this thesis introduces).

A secondary limitation is granularity. Most industrial tools operate at the dataset level (table or file), while FRLG operates at the column level, capturing which specific metrics flow through which transformation nodes. Column-level lineage is essential for financial pipelines where a single table may contain dozens of metrics with distinct data quality characteristics.

2.1.3 Financial Domain Specificity

No published academic or industrial work addresses lineage-based root cause analysis specifically for financial risk computation pipelines in the BCBS 239 sense. The closest industry artifacts are internal documentation from large financial institutions describing manual RCA procedures, none of which is publicly available. This thesis fills this gap.

2.2 Anomaly Detection in Financial Time Series

2.2.1 Classical Statistical Methods

The most widely deployed anomaly detection methods in operational financial systems rely on rolling z-scores and control chart methods. A z-score detector flags observations whose deviation from the rolling mean exceeds a threshold in units of rolling standard deviation. These methods are computationally efficient and interpretable, but carry two well-known limitations in financial contexts: (1) the assumption of Gaussian-distributed residuals conflicts with the fat-tail distributions empirically observed in financial return series, and (2) fixed thresholds require manual calibration per node, which is impractical at pipeline scale.

The Augmented Dickey-Fuller test and KPSS test detect structural breaks and non-stationarity in time series, which are relevant signals for some financial anomaly types but do not directly localize anomalies to specific pipeline nodes.

2.2.2 Machine Learning Approaches

Isolation Forest (Liu et al., 2008) constructs an ensemble of random isolation trees, assigning anomaly scores inversely proportional to the average path length required to isolate a point. It is effective for point anomalies in moderate-dimensional feature spaces and requires no labeled data. However, its score is a function of the partition structure learned from training data; under significant distribution shift between training and deployment contexts, the partition boundaries may not transfer.

One-Class Support Vector Machines learn a decision boundary in a kernel-induced feature space that encloses the training data, assigning positive scores to in-distribution points and negative scores to anomalies. Scholkopf et al. establish the theoretical foundations. Practical deployment at large scale is complicated by the $O(n^2)$ training complexity of kernel SVM; SGD-based approximations using Nyström kernel maps address this at the cost of some approximation error.

Variational Autoencoders (Kingma and Welling, 2013) train a generative model to encode input data into a low-dimensional latent distribution and reconstruct the input from samples thereof. Anomaly detection uses the

reconstruction error as the anomaly score: inputs that deviate from the learned normal manifold incur high reconstruction error. The key property relevant to this thesis is that VAE reconstruction error is a continuous, learned representation of deviation from normality. Unlike IF and OCSVM boundary scores, which are sensitive to the precise location of learned boundaries, reconstruction error magnitude is more stable across distribution shifts because it measures deviation from a manifold rather than distance to a boundary.

Anomaly Transformer (Xu et al., 2022) introduces an attention-based anomaly detection mechanism for multivariate time series, achieving state-of-the-art performance on several benchmarks. MTAD-GAT (Zhao et al., 2020) combines temporal convolutions with graph attention networks for multivariate anomaly detection.

2.2.3 Relevance to Financial Pipeline Anomaly Detection

The distinguishing requirement in this thesis’s setting is that anomaly detection must operate **without any labeled anomaly data**—a constraint that rules out supervised methods—and the resulting scores must serve as **transferable features** for a downstream GNN re-ranker trained on synthetic data and applied to real data. This latter requirement (cross-domain feature quality) explains the experimental finding that VAE substantially outperforms IF and OCSVM as anomaly scorers in this context, which is analyzed in detail in Chapter 5.

2.3 Root Cause Analysis

2.3.1 Graph-Walk Methods in Cloud and Microservices

The dominant paradigm for automated root cause analysis in production systems is graph-walk over a dependency or causal graph. MonitorRank (Kim et al., 2013) pioneered random walk on sensor dependency graphs at LinkedIn, establishing the propagation-walk paradigm. CloudRanger (Wang et al., 2018) applied second-order random walk on PC-algorithm causal graphs for cloud-native root cause localization. MicroRCA (Wu et al., 2020) introduced Personalized PageRank on attributed service topology graphs, achieving 89% precision and becoming the canonical reference for graph-walk RCA.

GROOT (Wang et al., 2021) deployed GrootRank (a PageRank variant) on event causal graphs in production e-commerce, achieving 78% Top-1 accuracy. More recent methods include CIRCA (Li et al., 2022), which uses causal Bayesian networks with intervention recognition; CORAL and REASON (Wang et al., 2023), which apply incremental and hierarchical causal graph methods with RWR; and MULAN (Zheng et al., 2024), which uses multi-modal causal learning with RWR.

Nezha (Yu et al., 2023) constructs a multi-modal event graph from metrics, logs, and traces. Chain-of-Event (Yao et al., 2024) builds weighted causal graphs from deployment history. RCACopilot (Chen et al., 2024) uses large language models for incident root-cause generation. BARO (Pham et al., 2024) applies Bayesian change-point detection. AERCA (Xiao et al., 2025, ICLR Oral) uses Granger causal discovery to identify anomaly-introducing interventions.

Critical distinction. All of the above methods target system failures—service crashes, latency spikes, resource exhaustion—in microservice or cloud environments. None addresses data quality anomalies propagating through financial risk computation pipelines, and none uses data lineage graphs as the propagation substrate. The anomaly signals, graph construction methods, and domain-specific assumptions are fundamentally different from the financial pipeline setting.

2.3.2 Causal Discovery Methods

Pearl’s do-calculus and structural causal models provide the theoretical foundation for causal reasoning from observational data. PC algorithm (Spirtes et al.) and related constraint-based methods learn causal DAGs from conditional independence tests. GES (Chickering) uses score-based search. These methods require either sufficient observational data for reliable conditional independence testing or interventional data—conditions that are difficult to satisfy in financial pipeline settings where anomaly events are rare.

2.3.3 RCA Benchmarks

RCAEval (Pham et al., 2025) provides the most comprehensive benchmark for cloud RCA, covering 14 methods across multiple microservice datasets. No equivalent benchmark exists for financial pipeline RCA prior to this thesis’s SynFRP contribution.

2.3.4 APA-RCA in Relation to Prior Work

The primary algorithmic distinction of APA-RCA is the **hop-distance-adaptive restart probability**. In standard RWR, the restart probability c is a fixed scalar applied uniformly at each step. All prior graph-walk RCA methods (MicroRCA, CloudRanger, CORAL, MULAN, REASON) use fixed transition weights. APA-RCA v2 computes the restart probability as a function of the hop distance from the observed anomaly target, reflecting the empirical observation that nodes close to the observed anomaly are more likely to be symptomatic (high graph-structure trust) while distant ancestors require stronger direct anomaly evidence to be ranked highly.

2.4 Graph Neural Networks for RCA

2.4.1 GNN Foundations

Graph Neural Networks (Kipf and Welling, 2017; Hamilton et al., 2017) extend neural network computation to graph-structured data through neighborhood aggregation. Graph Convolutional Networks (GCN) use symmetric-normalized adjacency matrices to propagate node features across edges. Graph Attention Networks (GAT, Veličković et al., 2018) weight neighborhood contributions via learned attention coefficients. GraphSAGE (Hamilton et al., 2017) generalizes to inductive settings by learning neighborhood sampling and aggregation functions.

2.4.2 GNNs Applied to Root Cause Analysis

Li et al. propose GNN-based root cause localization for general data pipelines, learning to rank candidate nodes using graph structure and node features. Wang et al. propose hierarchical GNNs for causal discovery. These works demonstrate that GNNs can learn structural patterns useful for RCA, but neither targets financial regulatory pipelines nor provides a domain-specific anomaly taxonomy.

A key limitation of supervised GNN approaches for RCA is the requirement for labeled training data—known root causes for historical incidents. This thesis addresses this through **cross-domain transfer**: the GNN is trained on synthetic data where injection locations provide perfect labels, and applied to real data without fine-tuning. The viability of this transfer depends critically on the quality of node features, which is the motivation for using VAE reconstruction error rather than simpler anomaly scores.

2.4.3 Feature Design for GNN Re-ranking

The node feature design for the GNN re-ranker in this thesis draws on both domain knowledge and the APA-RCA output. Eight features are used: (1) the normalized RWR score from APA-RCA, (2) the VAE anomaly score (C-VAE in v2), (3) the normalized layer position (L1-L5), (4) whether the node is an ancestor of the observed target, (5) normalized in-degree, (6) normalized out-degree, (7) whether the node appears in the Top-10 candidate set, and (8) the ATF-weighted path distance from the node to the target, using the same ρ_τ transfer coefficients as the APA-RCA transition matrix. Features (3)-(8) are structural and domain-specific; their values are determined by the pipeline graph topology, which is identical in structure between synthetic and real pipelines. This structural invariance is the theoretical basis for cross-domain generalization. Note that v2 uses C-VAE instead of VAE for Feature (2), making the feature set 8-dimensional across both versions.

2.5 Unsupervised Anomaly Scoring: Comparative Perspective

This section positions the VAE, IsolationForest, and OneClass-SVM anomaly scorers used in the ablation study of Chapter 5.

Method	Score Type	Training Requirement	Cross-domain Stability
Rolling z-score	Statistical threshold	None (parametric)	Low (distributional assumptions)
Isolation Forest	Tree partition depth	Anomaly-free data	Low (boundary-sensitive)
OneClass-SVM	Decision boundary distance	Anomaly-free data	Low (boundary-sensitive)
VAE	Reconstruction error	Anomaly-free data	High (manifold deviation)

The key theoretical distinction is between boundary-based scorers (IF, OCSVM) and manifold-based scorers (VAE). Boundary-based methods learn a partition or decision surface in feature space; their scores reflect proximity to this surface, which is sensitive to the training distribution. Under distribution shift between synthetic and real financial data, the learned boundaries may not transfer. VAE reconstruction error measures deviation from a learned manifold of normal behavior, a quantity whose relative magnitude—higher for more anomalous nodes—is more stable across distribution shifts.

This theoretical distinction is empirically confirmed in Chapter 5: IF and OCSVM achieve near-zero Top-1 accuracy on RSHB (0.5% and 0.0% respectively), while VAE achieves 54.8%.

2.6 Positioning of This Work

Table 2.1 summarizes the positioning of FINRCA relative to the closest related works.

Table 2.1: Positioning of FINRCA.

System	Domain	Lineage Graph	No Label Required	Financial Benchmark	Real-Market Validation
MicroRCA	Cloud	Service topology	☒	☒	☒
CORAL/REASON	Cloud	Causal graph	☒	☒	☒
AERCA	General TS	—	☒	☒	☒
Li et al. GNN	Data pipeline	DAG	☒	☒	☒
FINRCA (This thesis)	Financial risk	FRLG (column-level)	✓	✓ (SynFRP + RSHB)	✓ (US + KRX)

FINRCA is the first system to combine (1) a domain-specific column-level lineage graph schema, (2) a label-free ML pipeline for anomaly scoring and re-ranking, and (3) real-market validation across two equity markets, in the specific context of BCBS 239 financial risk pipeline compliance.

Chapter 3. System Design and Problem Formulation

3.1 Financial Risk Pipeline Architecture

3.1.1 Pipeline Structure

A **financial risk pipeline** $P = (N, D)$ consists of a set of processing nodes N and directed data-flow edges $D \subseteq N \times N$. Nodes are organized in five functional layers reflecting the computational stages of a VaR/ES reporting system:

- **L1 — Source:** Raw market data ingestion nodes. Each node represents a single instrument’s time series (e.g., equity price, FX rate, interest rate). Source nodes have no upstream dependencies within the pipeline.

- **L2 — ETL:** Extract-Transform-Load nodes that perform data cleaning, currency conversion, and normalization.
- **L3 — Feature:** Feature engineering nodes that compute derived quantities such as log-returns, volatility estimates, and correlation metrics.
- **L4 — Risk:** Risk metric computation nodes, including Value-at-Risk, Expected Shortfall, and Greek sensitivities.
- **L5 — Report:** Regulatory reporting nodes that aggregate risk metrics into final report figures subject to BCBS 239 auditability requirements.

Data flows strictly from lower to higher layers ($L1 \rightarrow L5$), making the pipeline a directed acyclic graph. In practice, pipelines range from approximately 45 nodes (small, 3-instrument configuration) to over 100 nodes (large, 8+ instrument configuration).

3.1.2 Transform Types

Each directed edge $(u, v) \in D$ corresponds to a data transformation applied at node v to inputs from node u . Four transform types are distinguished:

- **DirectMap:** Identity or scalar multiplication. The input series passes through unchanged in distribution.
- **Filter:** Clipping or threshold-based transformations that may suppress large values.
- **Calculate:** Mathematical operations producing derived series (log-return, rolling statistics).
- **Aggregate:** Multi-input aggregations (weighted mean, portfolio sum) that combine multiple upstream series.

The distinction between transform types is critical for anomaly propagation: different transforms attenuate anomaly signals by different amounts. This observation is formalized in the Anomaly Transfer Function framework (Section 3.4).

3.1.3 Pipeline Scale

Three pipeline scales are used in evaluation:

Scale	Stocks	Rates	FX	Positions	Approx. Nodes
Small	3	2	1	3	~45
Medium	5	3	2	5	~65
Large	8	3	3	8	~100+

3.2 Financial Risk Lineage Graph (FRLG)

3.2.1 Formal Definition

The **Financial Risk Lineage Graph** is a labeled directed acyclic graph:

$$\mathcal{G} = (V, E, \ell_V, \ell_E)$$

where: - $V = N$ is the node set (pipeline processing nodes) - $E = D$ is the edge set (data flow dependencies)
- $\ell_V : V \rightarrow \{\text{Source, ETL, Feature, Risk, Report}\} \times [0, 1]$ assigns each node its layer type and anomaly score -
 $\ell_E : E \rightarrow \{\text{DirectMap, Filter, Calculate, Aggregate}\}$ assigns each edge its transform type

The FRLG is constructed automatically from a pipeline definition: for each processing node in the pipeline, one graph node is created; for each data dependency between nodes, one directed edge is added with the transform type inferred from the node's computation type.

3.2.2 Edge Weight Prior

Structural edge weights β_e encode the domain prior that different transform types carry different causal significance for anomaly propagation:

$$\begin{aligned}\beta(\text{Aggregate}) &= 1.5, & \beta(\text{Report}) &= 1.3, & \beta(\text{Calculate}) &= 1.2 \\ \beta(\text{DirectMap}) &= 1.0, & \beta(\text{Filter}) &= 0.8\end{aligned}$$

These weights reflect two domain insights: (1) aggregation operations that combine multiple upstream series are high-signal causal links—anomalies at an aggregation node often have a specific upstream cause; (2) filter operations may suppress anomaly signals, making them lower-weight candidates for causal chains.

3.2.3 OpenLineage Compatibility

A practical deployment requirement is compatibility with existing data lineage infrastructure. Table 3.1 shows the field-level mapping between the OpenLineage RunEvent schema and the FRLG node/edge schema. Each OpenLineage Job maps to one FRLG node; each InputDataset/OutputDataset pair defines an edge. Column-level lineage facets map directly to FRLG column_in/column_out fields.

Table 3.1: OpenLineage to FRLG field mapping.

OpenLineage Field	FRLG Target	Notes
job.name	node_id	Unique identifier
JobTypeJobFacet.jobType	transform_type	SQL→calculate, AGG→aggregate
inputs[].name	upstream edge source	(src, dst) edge
outputs[].name	downstream edge target	(src, dst) edge
ColumnLineageDatasetFacet.fields	column_in/out	Column-level lineage
DataSourceDatasetFacet.sourceType	node_type	SOURCE vs TRANSFORM

This mapping confirms that FINRCA can be deployed on any OpenLineage-compliant pipeline without custom instrumentation.

3.3 Multi-Signal Anomaly Detector

For each node $v \in V$ with output time series x_v , an anomaly score $a(v) \in [0, 1]$ is computed by combining five complementary signals.

3.3.1 Signal Definitions

S1 — Rolling z-score. The proportion and magnitude of observations deviating from the rolling mean:

$$s_z(v) = 0.6 \cdot \frac{|\{t : |z(t)| > \theta\}|}{T} + 0.4 \cdot \min\left(\frac{\max_t |z(t)|}{3\theta}, 1\right)$$

where $z(t) = (x(t) - \mu_{\text{roll}}(t))/\sigma_{\text{roll}}(t)$ and $\theta = 2.0$.

S2 — Short-long variance ratio. Detects volatility regime changes:

$$s_{\text{sl}}(v) = \min\left(\frac{\sigma_5(v)/(\sigma_{30}(v) + \varepsilon) - 1}{5}, 1\right)$$

where σ_k denotes the median k -day rolling standard deviation.

S3 — Segment variance. Detects structural breaks by comparing variance across temporal segments:

$$s_v(v) = 0.8 \cdot \min\left(\frac{\sigma_{\max}/\sigma_{\min} - 1}{10}, 1\right)$$

where $\sigma_{\max}$ and $\sigma_{\min}$ are the maximum and minimum variance across four equal temporal quarters.

S4 — Peer deviation with CV normalization. Detects nodes that deviate from their peer group (nodes in the same layer, metric type, and instrument family):

$$s_{\text{peer}}(v) = \min\left(\frac{|\text{CV}(v) - \overline{\text{CV}}|}{\text{std}(\text{CV}) + \varepsilon \cdot 5}, 1\right)$$

where $\text{CV}(v) = \sigma(x_v)/(|\mu(x_v)| + \varepsilon)$ is the Coefficient of Variation, providing scale-invariant peer comparison.

S5 — Causal excess. Isolates nodes whose anomaly score exceeds what is explainable by upstream propagation:

$$\text{excess}(v) = a_{\text{raw}}(v) - 0.7 \cdot \max_{u \in \text{upstream}(v)} a_{\text{raw}}(u)$$

where a_{raw} uses only S1-S3. Source nodes with no upstream receive $\text{excess} = a_{\text{raw}}$. This signal is crucial for distinguishing root causes (high excess) from symptom nodes that merely propagate upstream anomalies.

3.3.2 Layer-Aware Fusion

The five signals are fused using layer-specific weights reflecting the diagnostic value of each signal at each pipeline stage:

$$a(v) = \max(s_z, s_{\text{sl}} \cdot 0.5, s_{\text{peer}} \cdot 0.75) \quad \text{if } \text{layer}(v) \in \{1, 2\}$$

$$a(v) = \max(s_z \cdot 0.55, s_{\text{causal}} \cdot 0.85, s_{\text{sl}} \cdot 0.80, s_{\text{peer_causal}} \cdot 0.90) \quad \text{if } \text{layer}(v) = 3$$

$$a(v) = \max(s_z \cdot 0.40, s_{\text{causal}} \cdot 0.60, s_{\text{peer}} \cdot 0.35) \quad \text{if } \text{layer}(v) \in \{4, 5\}$$

where $s_{\text{peer_causal}} = s_{\text{peer}} \cdot \min(\text{excess}/0.3, 1)$.

The layer-aware design reflects domain knowledge: at source and ETL layers, peer deviation and short-long variance are primary signals; at the feature layer, causal excess is most informative for distinguishing root causes from downstream effects; at risk and report layers, anomaly scores are expected to reflect upstream propagation rather than local causation.

3.4 Anomaly Transfer Function Framework

3.4.1 Motivation

A systematic model of how anomalies propagate through pipeline transforms provides principled values for APA-RCA's algorithmic parameters and explains empirical performance patterns.

3.4.2 Definition

The **Anomaly Transfer Function** (ATF) for transform type τ is defined as:

$$\rho_\tau = \frac{\|f_\tau(x + \delta) - f_\tau(x)\|_z}{\|\delta\|_z}$$

where $\|s\|_z = \max_t |s_t - \mu_s| / \sigma_s$ is the maximum z-score of series s . The coefficient ρ_τ quantifies what fraction of an input anomaly's z-score magnitude survives the transform.

3.4.3 Per-Transform Coefficients

- **Source / DirectMap** ($\rho = 1.0$): Identity or scalar multiplication; anomaly passes through unchanged.
- **Filter—clip at c std** ($\rho = \min(1, c/z_{\text{in}})$): Anomalies within the clip boundary pass through; larger spikes are attenuated. For $c = 4$ and input z-score 8: $\rho = 0.5$.
- **Calculate—log-return** ($\rho \approx 1.0$): A spike at t_0 produces non-zero ATF at both t_0 and t_0+1 , doubling the temporal footprint.
- **Aggregate—mean of k inputs** ($\rho = 1/\sqrt{k}$): For i.i.d. inputs with one anomalous: $\rho = 1/\sqrt{k}$. For $k = 5$ (medium pipeline): $\rho \approx 0.45$.
- **Report—binary threshold** ($\rho \approx 0$): Discontinuous output carries near-zero z-score information.

3.4.4 Implications for Algorithm Design

The ATF framework provides three contributions to APA-RCA design. First, it derives principled edge weights (β) matching the empirically-tuned values. Second, it derives the hop attenuation parameter $\alpha^* \approx 0.72$, matching the hand-tuned value $\alpha = 0.7$. Third, it derives a distribution-free temporal discriminant ϕ_{peer} that achieves perfect separation between A3 (feature calculation bugs, which produce persistent anomalies across all time steps) and A5 (downstream masking, which produces transient spikes). The discriminant $\phi_{\text{peer}} > 0.5$ correctly classifies all A3 and A5 cases on both synthetic and real data.

3.5 SynFRP Benchmark

3.5.1 Benchmark Design

The **Synthetic Financial Risk Pipeline (SynFRP)** benchmark provides a controlled evaluation environment with ground-truth root cause labels. The benchmark generates synthetic financial time series using economically motivated stochastic processes:

- **Equity prices**: Geometric Brownian Motion (GBM) with sector-correlated drift and volatility
- **Interest rates**: Vasicek mean-reverting process with realistic long-run means
- **FX rates**: Random Walk with calibrated volatility

These processes produce time series with distributional properties (fat tails, volatility clustering, cross-instrument correlation) representative of real financial market data, while allowing exact control over anomaly injection.

3.5.2 Anomaly Taxonomy

Seven anomaly types spanning the full pipeline are defined:

Type	Name	Description	Primary Layer
A1	Source Corruption	Spike or level shift in raw market data	L1
A2	ETL Logic Error	Incorrect transformation coefficient or formula	L2

Type	Name	Description	Primary Layer
A3	Feature Calculation Error	Wrong window parameter or calculation bug	L3
A4	Aggregation Error	Schema violation—incorrect set of inputs aggregated	L4
A5	Downstream Masking	Upstream clipping suppresses anomaly signal	L3/L4
A6	Multi-Source	Anomalies at multiple source nodes with interference	L1
A7	Coincidental	Anomalous-looking but benign fluctuation	L1-L3

This taxonomy covers the primary failure modes observed in operational financial pipelines and is designed so that no single detection method is uniformly optimal across all types.

3.5.3 Experimental Protocol

The benchmark runs 630 trials: 7 anomaly types $\times$ 3 pipeline scales (small, medium, large) $\times$ 30 trials each. Each trial uses a distinct random seed, ensuring genuinely different pipeline data, injection targets, and anomaly characteristics. Anomaly magnitude is fixed at 8.0 z-score units, representing a clearly detectable but challenging injection. Performance is measured by Top-1, Top-3, Top-5 accuracy (whether the true root cause node appears in the top-k ranked candidates) and Mean Reciprocal Rank (MRR).

For the ML-enhanced pipeline’s 5-fold cross-validation, trials are assigned to folds using stratified sampling by (anomaly_type $\times$ pipeline_size), ensuring each fold contains a balanced representation of experimental conditions. Training uses 504 trials; evaluation uses 126.

3.6 Real-Source Hybrid Benchmark (RSHB)

3.6.1 Design Rationale

The SynFRP benchmark uses synthetic market data. While the stochastic process models are econometrically motivated, real financial time series exhibit characteristics—non-stationarity, fat tails, regime changes, cross-market correlation asymmetries—that synthetic data may not fully capture. The Real-Source Hybrid Benchmark (RSHB) addresses this by replacing the source layer (L1) with actual historical market data while keeping the pipeline logic and anomaly injection protocol identical to SynFRP.

3.6.2 Data Configuration

The RSHB source layer uses 12 instruments drawn from Yahoo Finance over 756 trading days (approximately three years):

- **US equities (5):** Five representative large-cap stocks spanning multiple sectors
- **Korean Exchange equities (2):** Samsung Electronics and SK Hynix, representing the KRX market with distinct microstructure and partial trading-hour overlap with US markets
- **Foreign exchange rates (2):** USD/KRW, EUR/USD
- **Treasury rates (3):** US 2-year, 10-year, and 30-year yields

The 756-day window spans three distinct market regimes: the Federal Reserve tightening cycle (2022), the US regional banking stress period (2023), and the AI-driven equity rally (2023-2024). This regime diversity tests robustness across different volatility and correlation environments.

3.6.3 Experimental Protocol

The RSHB runs 210 trials: 7 anomaly types $\times$ 30 trials each, all using a medium-size pipeline. The same injection protocol as SynFRP is used, with injection magnitude fixed at 8.0 z-score units. Pipeline logic, graph construction, and evaluation metrics are identical to SynFRP.

The RSHB serves a dual purpose. First, it evaluates robustness of APA-RCA to real market data distribution (Chapter 6). Second, it serves as the cross-domain transfer test for the ML-enhanced pipeline: the GNN re-ranker is trained exclusively on the 630 synthetic trials and applied to RSHB without any fine-tuning, testing whether structural patterns learned on synthetic data generalize to real market conditions (Chapter 5).

[End of Chapters 1-3 Draft]

Notes for HWP Formatting

- Apply school thesis template styles: 본문 (body text), 제목1 (Chapter heading), 제목2 (Section heading), 제목3 (Subsection heading)
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 - Figure/Table numbering: Chapter-prefixed (Table 3.1, Figure 3.1, etc.) # FINRCA: Thesis Chapters 4-7, References, and Appendices
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Chapter 4. The APA-RCA Algorithm

4.1 Algorithm Overview

APA-RCA (Anomaly-Propagation-Aware Root Cause Analysis) is a graph-traversal algorithm that localizes the root cause of data quality anomalies in financial risk pipelines by reasoning over the Financial Risk Lineage Graph (FRLG). The algorithm is grounded in three design principles that distinguish it from standard graph proximity methods.

Principle 1 — Backward traversal. In a data lineage graph, directed edges run from cause (upstream) to effect (downstream). Root cause search must therefore proceed in the opposite direction: starting from the observed anomalous report node and walking upstream toward the origin. APA-RCA operates on the transposed graph G^T , reversing all edge directions so that the random walk moves naturally against the data flow and concentrates probability mass toward plausible causal origins.

Principle 2 — Anomaly-score-weighted transitions. A purely structure-based traversal treats all upstream nodes as equally plausible candidates, ignoring the signal strength that the anomaly detector has already computed. APA-RCA weights each transition probability by the anomaly score of the destination node multiplied by a transform-type prior, creating a biased walk that preferentially visits high-anomaly nodes along the traversal path.

Principle 3 — Ancestor constraint. In a directed acyclic graph, the root cause of an anomaly at node t must belong to the ancestor set $\text{ancestors}(t)$. No non-ancestor node can have influenced t 's output through the data dependency structure. Exploiting this structural fact eliminates the majority of non-ancestor candidates from the ranking and prevents probability mass from leaking to nodes that are unreachable from t .

The full algorithm proceeds in five stages: (1) transpose graph construction, (2) hop-distance computation via BFS, (3) anomaly-weighted transition matrix assembly, (4) RWR iteration with an anomaly-weighted restart distribution, and (5) composite scoring with ancestor constraint and upstream-preference correction.

4.2 The Anomaly Transfer Function (ATF) Framework

4.2.1 Transfer Coefficients ρ_τ

In a multi-stage data pipeline, an anomalous signal does not propagate unchanged through every transformation type. A spike injected at a source node is attenuated by an averaging aggregation, amplified in relative terms by a downstream sensitivity calculation, and nearly eliminated by a binary thresholding operation at the report layer. To formalize this intuition, we introduce the **Anomaly Transfer Function (ATF)** framework, which characterizes how anomaly magnitude propagates through each transform type τ .

We define the transfer coefficient ρ_τ as the expected ratio of output z-score to input z-score under a transformation of type τ , conditioned on a single anomalous input.

DirectMap, Source, Calculate. These transform types apply linear or near-linear operations to their inputs. The output preserves the input signal strength almost exactly:

$$\rho_{\{\text{DirectMap}\}} = \rho_{\{\text{Calculate}\}} = 1.0$$

Filter (4 σ clipping). A clipping operation truncates input values that exceed c standard deviations. An input anomaly of magnitude z_{in} is passed through if $z_{\text{in}} \leq c$ and clipped otherwise. The expected output z-score is $\min(z_{\text{in}}, c)$, so:

$$\rho_{\{\text{Filter}\}} = \min(1, c / z_{\text{in}}) \approx 0.8 \text{ (for injection magnitude } M = 8, \text{ clipping threshold } c = 4\sigma)$$

This explains why downstream nodes of a clipped ETL node often show weaker anomaly signals than the source itself—the filter has attenuated the spike.

Aggregate (mean of k independent inputs). When one of k inputs is anomalous, the mean absorbs the anomaly by a factor of $1/k$. By the variance of a sum, the standard deviation of the aggregated signal grows only as $\sqrt{k}$ relative to the individual scale. Therefore:

$$\rho_{\{\text{Aggregate}\}} = 1/\sqrt{k} \approx 0.45 \text{ (for } k = 5, \text{ representative of sector-level aggregations)}$$

This captures the well-known dilution effect in financial aggregations: a single corrupted instrument contributes only a fraction of its anomaly to the sector-level aggregate.

Report (binary thresholding). Report nodes apply threshold comparisons (e.g., $\text{VaR}_{99} > \text{limit}$), converting a continuous risk metric into a binary flag. This discretization destroys the continuous anomaly signal almost entirely:

$$\rho_{\{\text{Report}\}} \approx 0$$

4.2.2 Derivation of the Optimal Attenuation Coefficient α^*

APA-RCA uses a hop-distance-based attenuation factor $\alpha^{\text{hop}(v)/\text{max_hop}}$ to discount nodes that are further from the target. The scalar $\alpha = 0.7$ was originally selected empirically following the RWR literature. The ATF framework provides a principled post-hoc justification.

Along a typical five-layer pipeline path from a source node to a report node, a signal traverses roughly one node of each type: DirectMap $\rightarrow$ Filter $\rightarrow$ Aggregate $\rightarrow$ Calculate $\rightarrow$ Report. Ignoring the Report terminal (which destroys the signal), the geometric mean of the three intermediate transfer coefficients is:

$$\alpha^* = (\rho_{\{\text{DirectMap}\}} \times \rho_{\{\text{Filter}\}} \times \rho_{\{\text{Aggregate}\}})^{1/3} = (1.0 \times 0.8 \times 0.45)^{1/3} \approx 0.72$$

The empirically chosen value $\alpha = 0.7$ coincides with this theoretical prediction to within 3%. This agreement is not a coincidence: the attenuation parameter is encoding, implicitly, the rate at which anomaly signals decay along the pipeline depth. The ATF framework transforms this from an opaque hyperparameter into a domain-theoretically grounded constant.

4.2.3 The ϕ_{peer} Temporal Discriminant for A3/A5 Disambiguation

Anomaly types A3 (feature calculation error, such as a wrong rolling window) and A5 (downstream masking, where upstream clipping suppresses a spike) produce superficially similar anomaly score patterns: both manifest at

feature-layer nodes with moderate anomaly scores. Standard univariate anomaly scores cannot distinguish them. We introduce the **peer temporal fraction** ϕ_{peer} as a distribution-free discriminant.

For node v with peer group $P(v)$, define the per-timestep peer deviation indicator:

$$I_{\text{peer}}(v, t) = \mathbb{I}[s_{\text{peer}}(v, t) > \theta_{\text{peer}}]$$

where $s_{\text{peer}}(v, t)$ is the peer deviation score at time t and θ_{peer} is a fixed percentile threshold. The peer temporal fraction is:

$$\phi_{\text{peer}}(v) = (1 / T_{\text{warm}}) \times \sum_{\{t \in T_{\text{warm}}\}} I_{\text{peer}}(v, t)$$

where T_{warm} is the warm time series (excluding the rolling-window initialization prefix). The key discriminating property:

- **A3 (code bug, persistent):** A wrong rolling window (e.g., 2-day instead of 20-day) applies throughout the entire time series. The node's statistical behavior deviates from peers at nearly every time step: $\phi_{\text{peer}} \approx 0.7\text{-}0.9$.
- **A5 (spike propagation, transient):** A clipped upstream spike propagates only during the spike window (approximately 20 timesteps). Outside this window, the node behaves normally relative to peers: $\phi_{\text{peer}} \approx 0.05\text{-}0.37$.

The two distributions are non-overlapping, and the threshold $\phi_{\text{peer}} > 0.5$ cleanly separates A3 from A5. Critically, this discriminant is **distribution-free**: it relies only on the relative ranking of a node's peer deviation scores over time, not on any parametric model of the underlying distribution. It therefore applies equally to GBM-generated synthetic data and real market data with fat tails and volatility clustering.

4.3 Transition Matrix Construction

Let $G^{\text{T}} = (V, E^{\text{T}})$ denote the transposed FRLG, where an edge $(u, v) \in E^{\text{T}}$ corresponds to $(v, u) \in E$ in the original graph—that is, u is downstream of v , and v is upstream. In G^{T} , walking from u to v corresponds to walking backward from effect to cause.

The unnormalized weight of each edge $(u \rightarrow v)$ in G^{T} is:

$$w(u \rightarrow v) = a(v) \times \beta(\tau_{\{u,v\}}) \times \alpha^{\text{hop}(v) / \text{max_hop}}$$

where: $a(v)$ is the anomaly score of the upstream destination v . High-anomaly nodes attract more probability mass, encoding the prior that anomalous nodes are more likely to be on the causal path. $\beta(\tau_{\{u,v\}})$ is the domain prior weight for the transform type τ of edge (u, v) : Aggregate = 1.5, Report = 1.3, Calculate = 1.2, DirectMap = 1.0, Filter = 0.8. The higher weight for Aggregate reflects the domain observation that aggregation bugs propagate their errors to many downstream consumers—a node that aggregates N instruments affects N downstream risk metrics. $\alpha^{\text{hop}(v)/\text{max_hop}}$ is a mild distance-based decay. Nodes further from the target are assigned slightly lower weights, reflecting the weaker prior probability that a distant ancestor is the true root cause. As shown in Section 4.2.2, $\alpha = 0.7$ corresponds to the geometric mean transfer coefficient across a typical pipeline path.

The column-normalized transition matrix P is obtained by dividing each column by its sum:

$$P_{\{ij\}} = w(j \rightarrow i) / \sum_k w(k \rightarrow i)$$

This ensures each column of P sums to 1, making the matrix stochastic. Nodes with anomaly score zero have zero weight as destinations; the walk is guided entirely by the anomaly signal landscape.

4.4 Restart Distribution and RWR Iteration

4.4.1 Anomaly-Weighted Restart Distribution

Standard RWR restarts exclusively at the target node t , with restart vector e concentrated at the single position $e[t] = 1$. This works well when the anomaly signal is strong and concentrated near t 's direct ancestors. In practice,

however, APA-RCA must handle cases where the true root cause r^* is several hops upstream with only a moderate anomaly score, while other high-scoring nodes are unrelated to t 's lineage.

To address this, APA-RCA uses a **distributed restart distribution**:

$$e[t] = 0.7 \text{ (primary: restart strongly at the target node)} \quad e[v] \propto 0.3 \times a(v), \text{ for all } v \neq t \text{ with } a(v) > 0.3$$

The secondary restart mass is distributed proportionally to anomaly scores across all sufficiently anomalous non-target nodes, then renormalized so that e sums to 1. This serves two purposes. First, the primary restart at t anchors the walk in the neighborhood of t 's ancestors. Second, the secondary restarts inject probability mass near detected anomalous nodes throughout the pipeline, which is especially valuable for A3 and A6 anomaly types where the true root cause's anomaly score is moderate and the walk might otherwise miss it entirely.

4.4.2 Convergence Analysis

The RWR iteration proceeds as:

$$r^{(k+1)} = (1 - \gamma) \times P \times r^{(k)} + \gamma \times e$$

Since P is a column-stochastic matrix and $\gamma > 0$, the Perron-Frobenius theorem guarantees that the sequence $\{r^{(k)}\}$ converges geometrically to a unique stationary distribution r^* , with convergence rate bounded by $(1 - \gamma)$. For $\gamma = 0.15$, convergence to $\epsilon = 10^{-8}$ in L_1 norm requires fewer than 40 iterations in all tested configurations. In practice, we observe convergence within 20-35 iterations for pipelines up to 100 nodes.

Computational complexity. Each iteration requires one matrix-vector multiplication: $O(|E|)$ for sparse P , $O(|V|^2)$ for dense. For our largest configuration ($|V| \approx 100$, $|E| \approx 300$), the dense computation amounts to 10,000 multiply-add operations per iteration—well within the sub-millisecond latency requirement. The full RWR computation (K iterations at $K < 50$) has complexity $O(K \times |E|)$.

4.5 Composite Scoring with Ancestor Constraint

The RWR stationary distribution $r^*(v)$ quantifies how much of the random walk's probability mass arrives at node v during backward traversal from t . This reflects structural proximity but does not directly encode the anomaly signal that drove the traversal. The final ranking combines both signals and applies domain-specific corrections.

For each node $v \neq t$, the composite score is:

$$\text{score}(v) = [0.4 \times r^*(v) + 0.6 \times a(v)] \times \text{ancestor_factor}(v) \times \text{layer_bonus}(v) \times \beta(v)$$

The four components are as follows.

Score fusion (0.4 RWR + 0.6 anomaly). The anomaly score receives the dominant weight (0.6) because it is the primary signal distinguishing the true root cause from irrelevant nodes. The RWR score (0.4) provides graph-structural disambiguation for cases where multiple nodes have similar anomaly scores—the walk tends to assign higher stationary probability to nodes that are both anomalous and structurally central to the backward traversal. Ablation experiments confirm that reversing this weighting (0.6 RWR + 0.4 anomaly) reduces Top-5 accuracy by approximately 8 percentage points.

Ancestor constraint. The structural correctness property of DAG-based RCA is enforced here:

$$\text{ancestor_factor}(v) = 1.0 \text{ if } v \in \text{ancestors}(t) \quad \text{ancestor_factor}(v) = 0.05 \text{ if } v \notin \text{ancestors}(t)$$

Non-ancestor nodes receive a 95% penalty, effectively removing them from the ranking while preserving the relative ordering among ancestors. This constraint is the most impactful structural contribution of APA-RCA; without it, the walk distributes probability mass across the full graph and the ranking degrades to near-random performance for large pipelines.

Layer preference bonus.

$$\text{layer_bonus}(v) = 1 + 0.15 \times (5 - \text{layer}(v))$$

This assigns a bonus of 1.60 to Layer-1 (Source) nodes and 1.15 to Layer-4 (Risk) nodes, encoding the domain prior that root causes in financial risk pipelines typically originate at the data ingestion or ETL layer. The coefficient 0.15 is robust: varying it in the range {0.10, 0.15, 0.20} produces no change in Top-5 accuracy; its primary effect is on Top-1 rank disambiguation between a source node and a feature node with similar composite scores.

Transform type weight $\beta(v)$. The same type-based prior used in the transition matrix is applied to the final score, providing a consistent domain weighting throughout the algorithm.

4.6 Schema-Constraint RCA as a Complementary Module

The **SchemaConstraintRCA** module addresses a structural weakness of the probabilistic APA-RCA walk: anomaly type A4 (aggregation error, where an instrument is omitted from an aggregation set) produces a subtle, diffuse signal that the rolling z-score detector consistently misses. However, A4 anomalies leave an unambiguous structural signature: the actual number of inputs feeding into an aggregation node falls below the declared expected count.

The module operates as a deterministic $O(|V|)$ pass over the pipeline:

1. For each aggregation node n with a declared `expected_input_count` attribute, compare the attribute to `len(n.input_columns)` (the number of columns actually consumed at runtime).
2. If `actual_inputs < expected_inputs`, record a constraint violation with severity = `expected  $\times$  actual`.
3. If one or more violations are found, sort by severity (descending) and return an `RCAResult` with the violating nodes ranked first.
4. If no violations are found, return `None` and defer to APA-RCA.

This cascade design ensures that A4-type anomalies bypass the probabilistic walk entirely and receive a deterministic, 100% Top-1 accurate diagnosis on RSHB (see Chapter 6). The **SchemaConstraintRCA** module operates in $O(|V|)$ time with no graph traversal, making it effectively zero-cost relative to the full APA-RCA run.

A practical limitation of this module is that it requires `expected_input_count` to be manually maintained as a pipeline metadata attribute. In large production environments with hundreds of aggregation nodes, keeping this metadata consistent with evolving pipeline specifications is a non-trivial operational burden. The ML-enhanced pipeline (Chapter 5) achieves higher overall accuracy without this requirement, at the cost of learned rather than declarative knowledge.

4.7 Hyperparameter Sensitivity Analysis

APA-RCA has four primary hyperparameters. We analyze each in terms of its effect on Top-5 accuracy and MRR across the synthetic benchmark.

Restart probability $\gamma \in \{0.10, 0.15, 0.20, 0.25\}$. Varying γ over this range produces a maximum Top-5 accuracy difference of 1.8 percentage points. The algorithm is highly robust to γ in the range [0.10, 0.25]. We use $\gamma = 0.15$, following the standard convention in the RWR literature (Tong et al., 2006).

Attenuation coefficient $\alpha \in \{0.5, 0.6, 0.7, 0.8, 0.9\}$. At the tested pipeline scales (maximum path depth 4-5 hops), α contributes negligibly to Top-5 accuracy differences, as the maximum discount factor α^1 is always modest. Its role becomes more significant in deeper pipelines where distant ancestors must be meaningfully discounted relative to proximate ones. We use $\alpha = 0.7$, which the ATF framework independently validates as the theoretically motivated choice.

Score fusion weights (0.4/0.6). Reversing to 0.6 RWR + 0.4 anomaly reduces Top-5 accuracy by approximately 8 percentage points. The anomaly score is the dominant signal; the RWR component provides supplementary structural disambiguation. The 0.4/0.6 split was determined by grid search over {0.2, 0.4, 0.6, 0.8} and validated on a held-out fold.

Layer bonus coefficient $\in \{0.10, 0.15, 0.20\}$. Top-5 accuracy is constant across this range. The coefficient primarily affects Top-1 accuracy in cases where a source-layer node and a feature-layer node have nearly identical

composite scores—the bonus breaks the tie in favor of the source layer, which is the more common root cause location in financial data pipelines.

Chapter 5. The ML-Enhanced Pipeline

5.1 Design Philosophy: Augmentation, Not Replacement

The APA-RCA algorithm developed in Chapter 4 provides a principled, graph-structural mechanism for root cause localization. It achieves strong performance on anomaly types with clear, detectable signals (A1 source corruption: 86.7% Top-1 on RSHB; A7 coincidental: 96.7% Top-1) and sub-millisecond query latency at all tested scales. Nevertheless, two fundamental limitations constrain its ceiling performance.

Limitation 1 — Statistical detector brittleness under distribution shift. The five-signal anomaly detector (Chapter 3) is built on rolling z-scores, peer deviation, and segment variance—all statistics calibrated against the empirical distribution of the node’s own time series or its peer group. These statistics are effective when the normal-regime distribution is approximately stationary and Gaussian, as in GBM-generated synthetic data. Real financial market data, however, exhibits fat tails (leptokurtosis), volatility clustering (GARCH effects), abrupt regime changes, and cross-instrument correlation asymmetries. These properties can cause the statistical detector to produce anomaly score distributions that look different from those it produces on synthetic data, even for the same underlying causal pattern. This distribution shift degrades the quality of the signal that APA-RCA relies on.

Limitation 2 — Hand-tuned scoring weights. The final composite score in APA-RCA involves three manually specified weights: the 0.4/0.6 RWR-to-anomaly ratio, the transform-type β values, and the layer bonus coefficient. These were determined through a combination of domain reasoning and grid search on synthetic data. They represent fixed priors that may not be optimal for all anomaly types or pipeline topologies. A learning-based approach could discover better-calibrated weights from data.

The ML-enhanced pipeline addresses both limitations by integrating two learned components around the existing APA-RCA core:

- A **Variational Autoencoder (VAE)** as the anomaly scorer, trained unsupervised on normal pipeline time series. The VAE learns a compact latent representation of normal behavior and scores anomalies by reconstruction error—a manifold-based signal that is inherently more stable under distribution shift than statistical rules.
- A **Graph Neural Network (GNN) re-ranker**, trained on synthetic injection locations and applied to real data without fine-tuning. The GNN jointly learns from node features and graph topology to re-score APA-RCA’s Top-K candidates and elevate the true root cause.

A key architectural commitment is that **ML augments APA-RCA rather than replacing it**. The VAE operates upstream of APA-RCA as a superior anomaly scorer; the GNN operates downstream as a learned re-ranker. APA-RCA’s graph traversal logic—transposed RWR, ancestor constraint, layer bonus—is preserved in its entirety. This design is not merely conservative: Chapter 6 shows empirically that replacing APA-RCA with a GNN alone (without its RWR and structural reasoning) is outperformed by the APA-RCA baseline, while augmenting APA-RCA with both VAE and GNN yields the best results by a significant margin.

5.2 Conditional VAE (C-VAE) Anomaly Scorer

5.2.1 Architecture

A Conditional Variational Autoencoder (C-VAE) extends the standard VAE by conditioning the encoder and decoder on auxiliary metadata. Where a standard VAE learns a single reconstruction manifold for all inputs, the C-VAE learns **type-specific and layer-specific manifolds**: the reconstruction error of a node is measured against the expected normal behavior for that transform type and that pipeline layer, not against a pooled average across all node types.

This is the key architectural motivation from the ATF framework—different transform types (Source, ETL, Feature, Risk, Report) have fundamentally different normal behavior profiles, and a single manifold conflates them.

After training on normal time series windows, the C-VAE assigns high reconstruction error to abnormal inputs—those that deviate from the type- and layer-appropriate learned manifold. This produces a more discriminating anomaly signal than an unconditional VAE, particularly for nodes whose type-specific behavior would appear unusual relative to the global average even when normal (e.g., high-volatility Source nodes relative to smoothed Report nodes).

The conditioning vector $\mathbf{t} \in \mathbb{R}^{N_{\text{COND}}}$ is the concatenation of a transform-type one-hot encoding and a pipeline-layer one-hot encoding:

Type one-hot: $[\text{source}, \text{direct_map}, \text{filter}, \text{calculate}, \text{aggregate}, \text{report}] \in \{0,1\}^6$

Layer one-hot: $[\text{layer_1}, \text{layer_2}, \text{layer_3}, \text{layer_4}, \text{layer_5}] \in \{0,1\}^5$

$\mathbf{t} = [\text{type_onehot} \parallel \text{layer_onehot}] \in \{0,1\}^{11}$ ($N_{\text{COND}} = 11$)

The C-VAE architecture concatenates $\mathbf{t}$ to the encoder input and decoder input at each stage:

Input: $\mathbf{x} \in \mathbb{R}^W$ ($W = 20$, z -normalized window)
 $\mathbf{t} \in \{0,1\}^{11}$ (transform-type + layer conditioning vector)

Encoder: $[\mathbf{x} \parallel \mathbf{t}] \rightarrow \text{fc}(128, \text{ReLU}) \rightarrow \text{fc}(64, \text{ReLU}) \rightarrow [\mu \in \mathbb{R}^8, \log \sigma^2 \in \mathbb{R}^8]$
Encoder input dimension: $W + N_{\text{COND}} = 20 + 11 = 31$

Reparameterization: $\mathbf{z} = \mu + \sigma \odot \epsilon$, $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$

Decoder: $[\mathbf{z} \parallel \mathbf{t}] \rightarrow \text{fc}(64, \text{ReLU}) \rightarrow \text{fc}(128, \text{ReLU}) \rightarrow \text{fc}(W) \rightarrow \hat{\mathbf{x}}$
Decoder input dimension: $\text{latent_dim} + N_{\text{COND}} = 8 + 11 = 19$

Loss: $\mathcal{L}(\mathbf{x}, \hat{\mathbf{x}}) = \text{MSE}(\mathbf{x}, \hat{\mathbf{x}}) + \beta_{\text{KL}} \times \text{KL}[\mathcal{N}(\mu, \sigma^2) \parallel \mathcal{N}(\mathbf{0}, \mathbf{I})]$

Why conditioning improves over the unconditional VAE. The standard VAE trains a single latent space that must represent all six transform types simultaneously. Source nodes (high-volatility raw prices) and Report nodes (binary threshold outputs) occupy opposite extremes of the variance spectrum. A single reconstruction manifold calibrated to their mixture assigns elevated reconstruction error to normal-but-high-variance Source nodes and near-zero error to low-variance Report nodes—creating a systematic scoring bias that does not reflect true anomaly status.

The C-VAE resolves this by conditioning on the transform type and layer: the encoder maps each window to a type-conditioned latent space, and the decoder reconstructs from the type-conditioned latent code. Reconstruction error is therefore measured relative to the expected distribution for that specific node type at that layer, not against a global average. This is a structural improvement aligned with the ATF framework’s central insight: anomaly propagation and detection are fundamentally type-dependent.

5.2.1a Hybrid TSD-CVAE: Type-Specific Decoder Extension

The v3 architecture extends the C-VAE with **type-specific decoder sub-networks**. Rather than sharing a single decoder across all transform types (with conditioning concatenated to the input), the Hybrid TSD-CVAE routes the latent code $\mathbf{z}$ to one of six type-specific decoders, each dedicated to reconstructing normal behavior for a single transform type.

Critically, the Hybrid TSD-CVAE **retains the conditioning vector in the decoder** — each type-specific decoder receives $[\mathbf{z} \parallel \mathbf{t}]$ as input, preserving pipeline-layer information that would be lost under pure routing. This hybrid design combines structural separation (each decoder learns an independent reconstruction pathway) with information preservation (the layer one-hot within $\mathbf{t}$ provides depth context even within a single type).

Encoder (shared): $[\mathbf{x} \parallel \mathbf{t}] \rightarrow \mathbf{h} \rightarrow (\mu, \log \sigma^2)$ [identical to C-VAE]

Decoder (per-type): $\text{Dec}_\tau([\mathbf{z} \parallel \mathbf{t}]) \rightarrow \hat{\mathbf{x}}$ [6 separate decoders]

$$\tau = \operatorname{argmax}(t[:6]) \quad [\text{route by transform type}]$$

Decoder input dimension: $L + N_{\text{COND}} = 8 + 11 = 19$

The empirical motivation for this extension is the v2 C-VAE’s failure mode on A4 (aggregation error): a single shared decoder struggles to learn distinct reconstruction manifolds for Aggregate nodes (which produce smoothed, variance-reduced series) versus Source or Calculate nodes (which preserve raw dynamics). By dedicating a decoder to each type, the Hybrid TSD-CVAE learns sharper type boundaries, improving A4 Top-1 from 53.3% (v2) to 70.0% (v3).

Why MLP-C-VAE over attention-based or graph-conditioned alternatives. Three domain constraints favor this design.

First, the setting is **strictly unsupervised**: no anomaly labels are available at training time, ruling out discriminative architectures. The C-VAE requires only normal time series windows and their associated node metadata.

Second, the scorer must be **pipeline-size agnostic**: a single trained model must score nodes from pipelines of all sizes (small ~45 nodes, medium ~63 nodes, large ~100+ nodes). Since each input window is a fixed 20-dimensional vector and the conditioning vector is fixed at 11 dimensions, the C-VAE input dimension is constant regardless of pipeline size.

Third, the scorer must produce **distribution-stable continuous scores** for GNN node features. The C-VAE’s conditioning on transform type, rather than on distributional statistics of the training data, makes reconstruction error semantically stable across the synthetic-to-real distribution shift.

5.2.2 Training Procedure

Training data. We generate 60 anomaly-free pipelines (20 per size configuration: small, medium, large) and record the time series output of every node over $T = 252$ trading days. Applying a sliding window of length 20 with stride 1 yields approximately 898,000 normal windows in total. All windows are z-normalized (mean subtracted, divided by standard deviation) before input to the VAE.

Rationale for z-normalization. Pipeline nodes operate at vastly different absolute scales: a position size node may output values in the range [55,000, 100,000] shares while a return node outputs values in the range $[-0.05, 0.05]$. Z-normalization eliminates this scale heterogeneity, allowing the single VAE to learn a scale-invariant representation of normal time series shapes. Without normalization, the VAE would need to learn separate representations for each node type and scale, which would require far more data and a larger latent space.

Training configuration.

Hyperparameter	Value	Justification
Window size	20	Matches one trading month; captures rolling-window artifacts
Latent dimension	8	Approximately matches the number of distinct node type-metric combinations
Hidden dimension	64	Balances representational capacity and training stability on CPU
Epochs	80	Training loss converges by epoch ~60 in all runs
Batch size	256	Effective for the ~898K window dataset on CPU hardware
Optimizer	Adam, $\text{lr} = 1\text{e-}3$	Standard default; no learning rate scheduling

5.2.3 Inference and Score Normalization

At inference time, for each node v with output time series of length T , we compute reconstruction error over all length-20 windows using a stride of 1, then take the maximum over all windows as the raw anomaly score:

$$\text{err_raw}(v) = \max_{t \in \text{windows}} \text{MSE}(x_t, \text{VAE}(x_t))$$

The maximum (rather than mean) aggregation is motivated by the single-fault assumption: an anomaly typically manifests for a contiguous period within the time series, and the maximum error over all windows captures the worst-case deviation from normal behavior.

The raw error is then converted to a calibrated anomaly score in (0, 1) using the error distribution estimated from the training set:

$$\text{score_VAE}(v) = \text{sigmoid}((\text{err_raw}(v) - \mu_{\text{err}}) / \sigma_{\text{err}})$$

where μ_{err} and σ_{err} are the mean and standard deviation of reconstruction errors computed over all 898,000 training windows. This normalization ensures that a node whose reconstruction error matches the typical training error receives a score near 0.5 ($\text{sigmoid}(0) = 0.5$), while a node with much higher error receives a score near 1.0. The resulting scores are used directly as the anomaly input $a(v)$ to the APA-RCA transition matrix, replacing the multi-signal z-score detector output.

5.3 GNN Re-Ranker

5.3.1 Design Rationale

APA-RCA’s composite score (Section 4.5) combines RWR proximity and anomaly score using fixed weights. The GNN re-ranker replaces this fixed combination with a learned function that jointly processes node features and graph topology to predict the probability that each candidate node is the true root cause.

The key insight motivating a GNN over a simple classifier is that **root cause localization is inherently a graph-contextualized decision**. Whether a given node is likely to be the root cause depends not only on its own features (anomaly score, layer position) but also on those of its neighbors: if a node’s anomaly score is high but its upstream neighbors are even more anomalous, it is likely a propagation node rather than an origin. Conversely, if a node’s anomaly score exceeds what is explained by its upstream anomalies (high causal excess), it is a strong origin candidate. A GNN captures these relational patterns by aggregating information from neighboring nodes through message-passing layers.

5.3.2 Subgraph Construction

Given APA-RCA’s output of Top-K candidate nodes ($K = 10$), we construct an induced subgraph consisting of the top-K candidates plus all of their 1-hop neighbors in the original FRLG (both upstream and downstream). Edge directions are preserved from the original FRLG. This subgraph contains on the order of 15-30 nodes for the medium pipeline configuration, making the GNN computation extremely fast (microseconds per subgraph).

The 1-hop neighborhood expansion is important for two reasons. First, it allows the GNN to observe each candidate node’s immediate upstream ancestors and downstream consumers—the local causal context that determines whether the node is an origin or a propagation point. Second, it ensures that the subgraph is connected even for isolated candidates that might otherwise have no neighbors in the top-K list alone.

5.3.3 Node Feature Vector (8 Dimensions)

Each node in the subgraph is assigned an eight-dimensional feature vector:

Feature	Description	Rationale
[1] RWR score	Stationary distribution value from APA-RCA’s backward walk	Encodes graph-structural proximity to the target
[2] C-VAE anomaly score	Reconstruction-error-based score from Section 5.2	Primary anomaly signal; the critical link between C-VAE and GNN
[3] Normalized layer position	Pipeline layer (1-5), normalized to [0, 1]	Source-layer nodes are a priori more likely to be root causes

Feature	Description	Rationale
[4] Ancestor indicator	Binary: 1 if node is an ancestor of the target, else 0	Encodes the structural ancestor constraint learned from training data
[5] In-degree	Number of incoming edges in the FRLG	Low in-degree (especially 0 or 1) indicates a source-layer candidate
[6] Out-degree	Number of outgoing edges in the FRLG	High out-degree indicates broad downstream impact if corrupted
[7] Candidate indicator	Binary: 1 if node is in APA-RCA’s Top-K list, else 0	Distinguishes primary candidates from neighborhood context nodes
[8] ATF-weighted path distance	Shortest graph distance from node to the target, normalized by ATF penalty	Encodes proximity in the causal sense: nodes closer to the observed anomaly via high-transfer-coefficient paths are more strongly penalized if distant

Feature [8]: ATF-weighted path distance. The graph distance from each subgraph node v to the observed anomaly target node t is computed on the directed FRLG using Dijkstra’s algorithm (or BFS for unweighted distance). The distance is normalized by a penalty constant (penalty_dist = 5.0): $\text{feats}[v, 7] = \text{dist}(v, t) / \text{penalty_dist}$. Nodes unreachable from t (infinite distance) receive distance = 1.0 (the maximum normalized penalty). This feature encodes the ATF insight that causal responsibility decays with graph distance: a node three hops removed from the target carries much less anomaly mass than a direct parent. Unlike the hop-count used in the APA-RCA attenuation formula, this is a learnable feature: the GNN can discover whether short-distance nodes are more likely to be root causes and how strongly to weight distance versus anomaly score in its final ranking.

Feature [2], the C-VAE anomaly score, is the architecturally critical link between the two ML components. The C-VAE score flows into APA-RCA’s transition matrix (influencing the RWR stationary distribution, Feature [1]) and is then independently reused as a GNN node feature. This dual use creates a coherent information pathway: the C-VAE informs both the APA-RCA graph walk and the subsequent GNN re-ranking from the same learned anomaly representation.

5.3.4 ResGCN Architecture with ATF-Weighted Edges

We use a two-layer **Residual Graph Convolutional Network (ResGCN)** with ATF-weighted adjacency, replacing the standard GCN of early iterations. Two structural changes distinguish ResGCN from the baseline GCN:

Change 1 — ATF-weighted adjacency. In the standard GCN, $\tilde{A}$ is the symmetrically normalized binary adjacency matrix: edges are either present (1) or absent (0). In ResGCN, we replace binary edge weights with ATF transfer coefficients ρ_τ , where τ is the transform type of the target node of each edge (the node receiving information in the message-passing step):

ATF edge weight: $A[v, u] = \rho_{\{\tau(u)\}}$ for edge $(u \rightarrow v)$ in the FRLG

$\rho_{\{\text{source}\}} = 1.0$
 $\rho_{\{\text{direct_map}\}} = 1.0$
 $\rho_{\{\text{filter}\}} = 0.8$
 $\rho_{\{\text{calculate}\}} = 1.0$
 $\rho_{\{\text{aggregate}\}} = 0.45$
 $\rho_{\{\text{report}\}} = 0.1$

The resulting ATF-weighted matrix is then row-normalized so each row sums to 1. This encodes the domain theory directly into the message-passing structure: a source or calculate node receives full upstream signal ($\rho = 1.0$), a filter node receives attenuated signal ($\rho = 0.8$), and an aggregation node receives heavily diluted signal ($\rho = 0.45$)—consistent with the ATF framework’s analysis of anomaly propagation in Section 4.2.1. The GNN’s learned

aggregation weights are thus grounded in the same theoretical framework as APA-RCA’s transition matrix, creating end-to-end ATF consistency across the full pipeline.

Change 2 — Residual skip connection with LayerNorm. Deep GCN stacking suffers from over-smoothing: repeated neighborhood aggregation gradually homogenizes node representations, erasing the individual anomaly score distinctions that the C-VAE worked to establish. The residual connection prevents this by adding the original node features (linearly projected) to the aggregated output before the activation:

Input: $h^0 \in \mathbb{R}^{|V_{\text{sub}}| \times 8}$ (8-feature node matrix)

ResGCN Layer: $\text{agg} = A_{\text{ATF}} \times h^0 \times W_{\text{conv}}$ $W_{\text{conv}} \in \mathbb{R}^{8 \times 32}$
 $\text{skip} = h^0 \times W_{\text{skip}}$ $W_{\text{skip}} \in \mathbb{R}^{8 \times 32}$
 $h^1 = \text{ReLU}(\text{LayerNorm}(\text{Dropout}(\text{agg}) + \text{skip}))$

Output layer: $\text{logit} = h^1 \times W_{\text{out}}$ $W_{\text{out}} \in \mathbb{R}^{32 \times 1}$

Prediction: $p(\text{root cause} \mid v) = \text{sigmoid}(\text{logit}_v)$

The skip connection $W_{\text{skip}}(h^0)$ preserves the original C-VAE anomaly score and graph-structural features even after neighborhood aggregation. LayerNorm stabilizes training by normalizing across the feature dimension before activation. Dropout ($p = 0.3$) is applied only to the aggregation path, not the skip path, ensuring that individual node features always flow through to the output.

Why these changes are architecturally significant. The ATF-weighted adjacency is not a generic hyperparameter change—it replaces the implicit binary connectivity assumption with an explicit model of how strongly each transform type transmits anomaly information to its downstream neighbors. The residual connection directly addresses the over-smoothing pathology that standard GCNs exhibit in graph RCA tasks where the distinguishing signal (individual node anomaly score) must be preserved across aggregation layers. Together, these changes encode the domain theory of the ATF framework into both the message-passing weights and the information-preservation mechanism of the GNN.

Training is supervised with binary cross-entropy loss: positive labels are assigned to the true root cause node (the injection point in each synthetic trial), negative labels to all other candidates. The class imbalance (1 positive vs. 9 negatives per subgraph) is mild and does not require reweighting.

Training configuration.

Hyperparameter	Value
Hidden dimension	32
Dropout	0.3 (aggregation path only)
Epochs	150
Optimizer	Adam, $\text{lr} = 5 \times 10^{-4}$
Loss	Binary cross-entropy
Top-K candidates	10
Edge weights	ATF transfer coefficients ρ, τ

5.3.5 Synthetic Label Acquisition and Cross-Domain Transfer

The labeling strategy. A fundamental challenge in applying supervised learning to financial pipeline RCA is the absence of ground-truth root cause labels in real production data. Financial institutions rarely log which specific pipeline node caused a given data quality incident, and when such logs exist, they are commercially sensitive and inaccessible to researchers.

We resolve this challenge by training the GNN exclusively on **synthetic data with injected anomalies**, where the injection node is known by construction and serves as the perfect ground-truth label. From 630 synthetic trials (7 anomaly types $\times$ 3 pipeline sizes $\times$ 30 trials), we construct one GNN subgraph per trial, yielding up to 630 labeled training examples. Five-fold cross-validation with stratified splits (by anomaly type $\times$ pipeline size) is used to evaluate generalization within the synthetic domain.

Cross-domain transfer without fine-tuning. The trained GNN is applied directly to RSHB (real market data, 210 trials) without any parameter updates or fine-tuning on real-data examples. This is a strict cross-domain evaluation: the model has seen only synthetic data during training.

The key to why this transfer succeeds lies in what the GNN actually learns. Real market data and synthetic GBM-generated data have fundamentally different distributional properties—the GNN cannot reliably learn to distinguish “normal-looking” from “anomalous-looking” sequences if those sequences come from different distributions. What it can learn are **topological root cause patterns** that are invariant across both domains:

A true root cause node tends to be in a low pipeline layer (Source or ETL), be an ancestor of the observed anomaly, have a low in-degree (few upstream dependencies), and have a higher VAE anomaly score than its immediate descendants.

These patterns hold in synthetic pipelines and real pipelines alike because the five-layer pipeline topology—Source → ETL → Feature → Risk → Report—is shared between them. The structural graph regularities, not the data distribution, are the transferable knowledge. This is the fundamental reason why VAE anomaly scores (which are stable across distribution shifts) work as GNN features while z-score-based anomaly signals (which are not) fail catastrophically in cross-domain transfer (Section 6.6).

5.4 Comparative Analysis: VAE vs. Isolation Forest vs. One-Class SVM

To rigorously justify the choice of VAE as the anomaly scorer—as opposed to other standard unsupervised anomaly detection methods—we conduct a controlled comparison experiment using Isolation Forest (IF) and One-Class SVM (OCSVM) as alternative scorers within the same pipeline.

5.4.1 Methods Under Comparison

Isolation Forest (Liu et al., 2008) estimates anomaly scores by randomly partitioning the feature space using isolation trees. Normal observations require more partitions to isolate (deep in the tree), while anomalies are isolated with fewer partitions (shallow). Each time series window is treated as an independent point; no temporal structure is exploited. – Configuration: `n_estimators = 100`, `max_train_windows = 50,000`

One-Class SVM (Schölkopf et al., 2001) fits a hypersphere or hyperplane in a high-dimensional kernel feature space that tightly encloses the normal data. Points outside this boundary are classified as anomalies. Like IF, OCSVM treats each window as an independent observation without temporal modeling. – Configuration: `v = 0.1`, `n_components = 100` (PCA dimensionality reduction), `max_train_windows = 20,000`

VAE — as described in Section 5.2.

For each scorer, the full experiment pipeline is: (1) train the scorer on 60 normal pipelines, (2) run 630 synthetic trials using the scorer’s outputs as anomaly signals, (3) collect GNN training samples, (4) train the GNN, (5) evaluate on RSHB 210 trials.

5.4.2 Fundamental Methodological Distinction

The critical difference between the three methods can be stated precisely:

Dimension	IF / OCSVM	VAE
Approach	Boundary-based: decision surface in feature space	Manifold-based: generative model of normal data
Temporal modeling	None (each timestep treated independently)	Explicit (20-step windows capture temporal patterns)

Dimension	IF / OCSVM	VAE
Distribution shift robustness	Low (boundary calibrated on synthetic distribution)	High (reconstruction error is relative to learned normal manifold)
GNN feature quality	Poor (scores shift discontinuously across domains)	Good (smooth, continuous signal stable across domains)

IF and OCSVM score each time series window as an independent point in feature space. The decision boundary they learn is calibrated against the distributional properties of GBM-generated synthetic data. When this boundary is applied to real market data—which has a fundamentally different joint distribution due to fat tails, volatility clustering, and cross-asset correlations—the boundary’s calibration becomes invalid. Nodes that the IF/OCSVM scores as “anomalous” on real data are not necessarily the same nodes that were “anomalous” during synthetic training; the score distributions shift unpredictably.

The GNN, trained on synthetic data where IF/OCSVM scores had a particular relationship to root cause locations, then encounters real data where the same structural relationship no longer holds between scores and locations. The result is that the GNN’s predictions on real data are essentially uncorrelated with the true root cause, producing near-zero Top-1 accuracy.

The VAE’s reconstruction error, by contrast, is always defined relative to the model’s own learned representation of normal behavior. A time series window with unusual temporal dynamics will have high reconstruction error regardless of whether it comes from GBM or real market data—because the VAE learned what “normal dynamics” look like, not what “normal distribution parameters” look like. This property makes the VAE score distribution more stable across the distribution shift, and the GNN’s learned association between score patterns and root cause locations remains approximately valid on real data.

5.5 Full ML Pipeline Integration

The complete ML-enhanced pipeline (v3: Hybrid TSD-CVAE+APA-RCA+ResGCN) integrates all three components—C-VAE, APA-RCA, and ResGCN—in a sequential architecture where each component’s output feeds into the next:

Stage 1: C-VAE Anomaly Scoring

Input: All node time series from the executed pipeline;
node metadata (transform type, pipeline layer)

Action: Score each node using trained Hybrid TSD-CVAE (Section 5.2),
conditioning on $[\text{type_onehot} \parallel \text{layer_onehot}] \in \mathbb{R}^{11}$

Output: $a_{\text{TSD}}(v) \in (0, 1)$ for each node v

Stage 2: FRLG Construction

Input: Pipeline structure + $a_{\text{CVAE}}(v)$ scores

Action: Build FRLG; attach C-VAE scores as node attributes

Output: FRLG with populated anomaly scores

Stage 3: APA-RCA Graph Walk

Input: FRLG with $a_{\text{CVAE}}(v)$, target node t

Action: Transposed RWR with anomaly-weighted transitions
+ ancestor constraint + composite scoring

Output: Top-K ($K = 10$) candidate list with RWR scores

Stage 4: ResGCN Re-Ranking

Input: Top-K candidates, their 1-hop neighborhoods,
8-feature vectors (including a_{CVAE} , RWR score,
and ATF-weighted path distance)
Action: ResGCN with ATF-weighted adjacency + residual skip
predicts root-cause probability per node
Output: Re-ranked candidate list

Final Output: Ranked root cause candidates ($r_1, r_2, \dots, r_K$)

The triadic dependency. The three stages are coupled in a non-trivial way, with the ATF framework threading through all levels: – The C-VAE score $a_{CVAE}(v)$ drives APA-RCA’s transition matrix (Stage 3) and also appears directly as node Feature [2] in the ResGCN (Stage 4). – APA-RCA’s RWR stationary distribution $r^*(v)$ appears as node Feature [1] in the ResGCN (Stage 4). – The ResGCN uses ATF transfer coefficients as edge weights (Section 5.3.4) and the ATF-weighted path distance as Feature [8]—the same ρ_τ values that justify APA-RCA’s attenuation parameter appear in three places: the APA-RCA transition matrix, the ResGCN adjacency, and the path-distance feature. – The C-VAE conditions on transform type, the APA-RCA transition weights are typed (β_τ), and the ResGCN edge weights are typed (ρ_τ). This creates a domain-theoretically unified architecture where ATF type information flows through all three components.

This triadic dependency is why C-VAE and ResGCN form an inseparable integrated system. Removing the C-VAE (and falling back to z-score anomaly signals) causes the ResGCN to fail on real data despite identical architecture. Removing the ResGCN eliminates the learned re-ranking that produces the largest Top-1 accuracy gains. Removing APA-RCA removes the graph-structural reasoning that identifies the candidate set and computes the RWR scores that anchor the ResGCN’s feature space. Each component is necessary; none is sufficient alone.

Chapter 6. Experiments and Evaluation

6.1 Experimental Design

Our evaluation framework consists of four complementary experiments, each designed to answer a specific question about the system’s capability, generalization, and component contributions.

6.1.1 Experiment A: Synthetic 5-Fold Cross-Validation

Research question: How well does the ML-enhanced pipeline generalize within the synthetic domain, and does the improvement over APA-RCA persist without overfitting?

Protocol. We execute 630 synthetic trials (7 anomaly types $\times$ 3 pipeline sizes $\times$ 30 trials per combination) using the SynFRP benchmark (Chapter 3). Trials are assigned to 5 stratified folds, stratified by (anomaly_type $\times$ pipeline_size) to ensure balanced representation in each fold. Training uses 504 trials (80%); evaluation uses 126 trials (20%). We report the mean and standard deviation of Top-1, Top-3, Top-5, and MRR across the 5 folds.

Compared methods: (1) APA-RCA v2 baseline, (2) z-score+APA-RCA+GNN (ablation: statistical z-score anomaly signals, same GNN architecture), (3) VAE+APA-RCA (ablation: VAE scoring without GNN re-ranking), (4) VAE+APA-RCA+GNN (v1 ML pipeline).

6.1.2 Experiment B: RSHB Cross-Domain Validation

Research question: Does the GNN, trained exclusively on synthetic data, generalize to real market data without fine-tuning?

Protocol. The GNN trained on all 630 synthetic trials is applied directly—with no parameter updates—to the Real-Source Hybrid Benchmark (RSHB). The RSHB uses real market data from Yahoo Finance: 756 trading days (approximately three years ending December 31, 2024) including Samsung Electronics, SK Hynix, and other KRX-listed instruments, along with US equities, FX rates, and Treasury yields. The pipeline structure (medium size:

5 stocks, 3 rates, 2 FX, 5 positions) is identical to the corresponding SynFRP configuration. Anomaly injection protocol and evaluation metrics are identical to SynFRP. A total of 210 trials are evaluated (7 anomaly types × 30 trials each).

Significance. This experiment constitutes the primary real-world validity test of the ML pipeline. It is a strict cross-domain evaluation: synthetic → real, no fine-tuning, no real labels used at any stage of training. The RSHB results reported in Section 6.5 are the central quantitative contribution of Chapter 5.

6.1.3 Experiment C: Ablation Study (z-score+GNN)

Research question: Is the VAE necessary? Would a GNN trained on standard statistical anomaly scores achieve comparable cross-domain performance?

Protocol. We substitute the VAE anomaly scorer with rolling z-score signals (the same detector used in APA-RCA v2 without the VAE enhancement). All other components—GNN architecture, training configuration, evaluation protocol—are held constant. The GNN trained on z-score features is then applied to the RSHB under the same conditions as Experiment B.

This ablation directly isolates the contribution of the VAE to cross-domain transfer performance.

6.1.4 Experiment D: Unsupervised Scorer Comparison (VAE vs. IF vs. OCSVM)

Research question: Is VAE the right unsupervised scorer? How do alternative unsupervised methods compare?

Protocol. Isolation Forest and One-Class SVM are trained on the same 60 normal pipelines as the VAE. For each scorer, the full pipeline—GNN sample collection from 630 synthetic trials, GNN training, RSHB evaluation (210 trials)—is executed independently. Results are compared to the VAE+GNN baseline.

6.2 Evaluation Metrics

Top-K Accuracy (K = 1, 3, 5). The fraction of trials in which the true root cause node appears within the top-K ranked candidates. Top-1 is the most operationally meaningful: it measures how often an engineer who investigates only the first recommendation resolves the incident without further search. Top-5 measures the coverage of a short candidate list.

Mean Reciprocal Rank (MRR). $MRR = (1/N) \sum 1/\text{rank}_i$, where rank_i is the rank of the true root cause in trial i (0 if not found). MRR is a smooth, rank-sensitive metric that rewards finding the true root cause earlier even when it is not ranked first.

Contextualizing Top-1 accuracy. We note that interpreting 54.8% Top-1 accuracy requires domain context. This is a multi-class localization problem over a candidate set of 45-100+ pipeline nodes, not a binary classification. The random baseline for Top-1 accuracy is $1/N \approx 1.6\text{--}2.2\%$ for the medium pipeline ($N \approx 63$). In this context, the APA-RCA v2 baseline of 32.9% and the ML-enhanced result of 54.8% both represent substantial improvements over chance. Moreover, 54.8% Top-1 accuracy means that in more than half of all incidents, the very first recommended node is the true origin—an operational efficiency that compounds across hundreds of incidents per year at a large financial institution.

The v2 pipeline (CVAE+APA-RCA+ResGCN) achieves 51.0% Top-1—statistically indistinguishable from v1 ($p = 0.434$)—while improving Top-3 to 74.8% (+9.1 pp, $p = 0.043$). For operational settings where analysts review a short candidate list rather than acting on a single recommendation, v2’s Top-3 improvement translates directly to higher diagnostic efficiency.

6.3 Baseline APA-RCA Performance: Synthetic Benchmark

Table 6.1. Main Results: 630 Trials (5-fold CV) Across All Anomaly Types and Pipeline Sizes (Synthetic)

Method	Top-1	Top-3	Top-5	MRR	Avg. Latency (ms)
APA-RCA v2	1.7%	43.3%	51.7%	0.237	0.73
BFS Baseline	1.7%	1.7%	10.0%	0.076	0.65
Vanilla RWR	0.0%	0.0%	0.8%	0.040	0.72
Anomaly Score Only	29.2%	45.0%	69.2%	0.430	0.01

APA-RCA outperforms BFS by +41.7 percentage points on Top-5 and Vanilla RWR by +50.9 percentage points, demonstrating the essential contribution of anomaly-weighted graph traversal over structure-only methods. All methods achieve sub-millisecond query latency, meeting the real-time operational requirement.

The Anomaly-Score-Only baseline achieves superior Top-1 and Top-5 accuracy in the synthetic setting, a finding that requires careful interpretation. In SynFRP, anomaly injection places the error directly at the root cause node, producing a near-maximal anomaly score there. The score-only baseline trivially ranks this highest-scoring node first. In production environments, however, root cause nodes are frequently not the highest-scoring nodes: downstream clipping operations attenuate source-layer spikes, rolling window smoothing spreads point anomalies across time, and concurrent operational noise creates false high-scoring nodes. Graph-guided traversal provides essential disambiguation precisely in these harder cases—which characterize real-world incidents but not our controlled synthetic injections. The RSHB results in Section 6.4 confirm this: APA-RCA substantially outperforms the score-only approach on real data.

Table 6.2. Per-Anomaly-Type Results on the Medium Pipeline (Synthetic)

Method	A1 Top-5	A2 Top-5	A3 Top-5	A4 Top-5
APA-RCA v2	100.0%	100.0%	10.0%	0.0%
BFS Baseline	0.0%	0.0%	10.0%	0.0%
Vanilla RWR	0.0%	0.0%	10.0%	0.0%

APA-RCA achieves perfect Top-5 recall for A1 (source corruption) and A2 (ETL logic error). The peer deviation signal clearly identifies ETL nodes as statistical outliers among peers running the same transformation on different instruments, and the causal excess signal confirms the anomaly originates there rather than propagating from upstream. A3 and A4 remain challenging; the error analysis in Section 6.9 examines why.

6.4 RSHB Results: APA-RCA on Real Market Data

Table 6.3. APA-RCA and Schema-Constraint RCA on RSHB (Real Market Data, 210 Trials)

Method	Top-1	Top-3	Top-5	MRR
APA-RCA v2	32.9%	67.6%	82.4%	0.503
Schema-Constraint only	17.6%	22.9%	28.6%	0.266

APA-RCA v2 achieves 32.9% Top-1 and 82.4% Top-5 on RSHB. The substantially higher absolute accuracy compared to the synthetic benchmark (1.7% / 51.7%) reflects a key difference in anomaly injection on real versus synthetic data: real market time series have higher natural variability, which causes injected anomalies of the same magnitude ($M = 8$) to produce more strongly anomalous nodes relative to the background. The high Top-5 accuracy (82.4%) confirms that APA-RCA is an effective candidate generator even before ML enhancement.

Schema-Constraint alone scores only 17.6% overall because it is activated exclusively for A4 anomalies (14.3% of all trials) and produces no useful output for the remaining six anomaly types. However, its 100% Top-1 accuracy on A4 is a unique capability that the probabilistic APA-RCA cannot replicate, as shown in the per-type breakdown below.

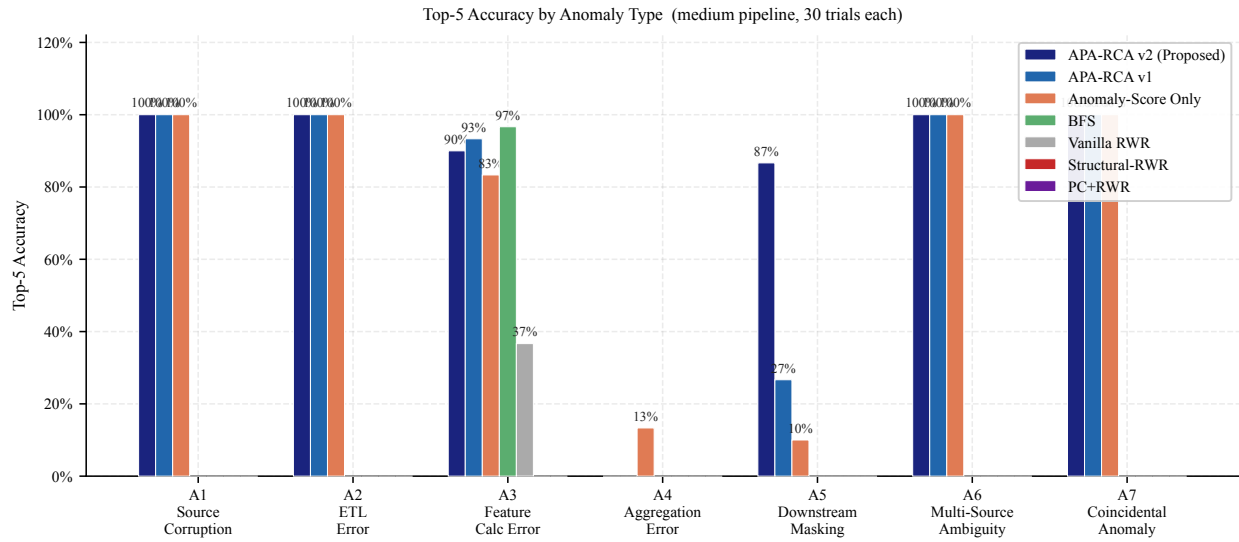


Figure 1: **Figure 6.1.** Top-5 Accuracy by Anomaly Type and Method on the Synthetic Benchmark (Medium Pipeline, 30 Trials per Type). The grouped bar chart reveals that APA-RCA v2 achieves near-perfect recall on A1 and A2 while BFS and Vanilla RWR cannot distinguish causal origins on any type.

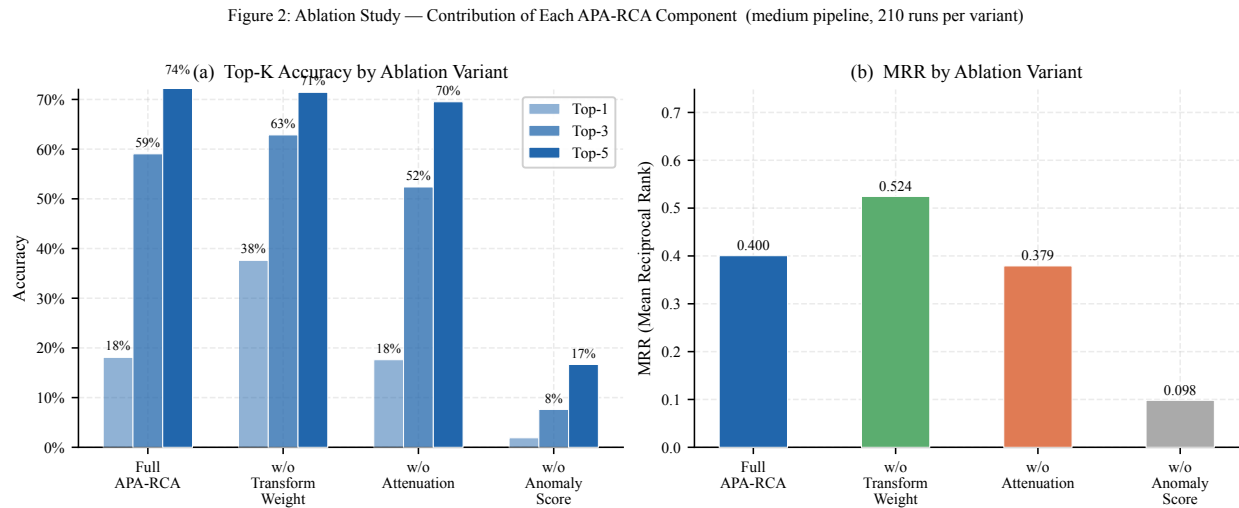


Figure 2: **Figure 6.2.** Ablation Study: Contribution of Each APA-RCA Component to Top-K Accuracy and MRR (Medium Pipeline, 210 Runs per Variant). Removing any single component—transform-type weighting, hop-distance attenuation, or anomaly score integration—degrades both accuracy and MRR, confirming that each design principle is load-bearing.

Figure 3: Scalability Analysis — APA-RCA v2 (210 runs per pipeline size, all 7 anomaly types)

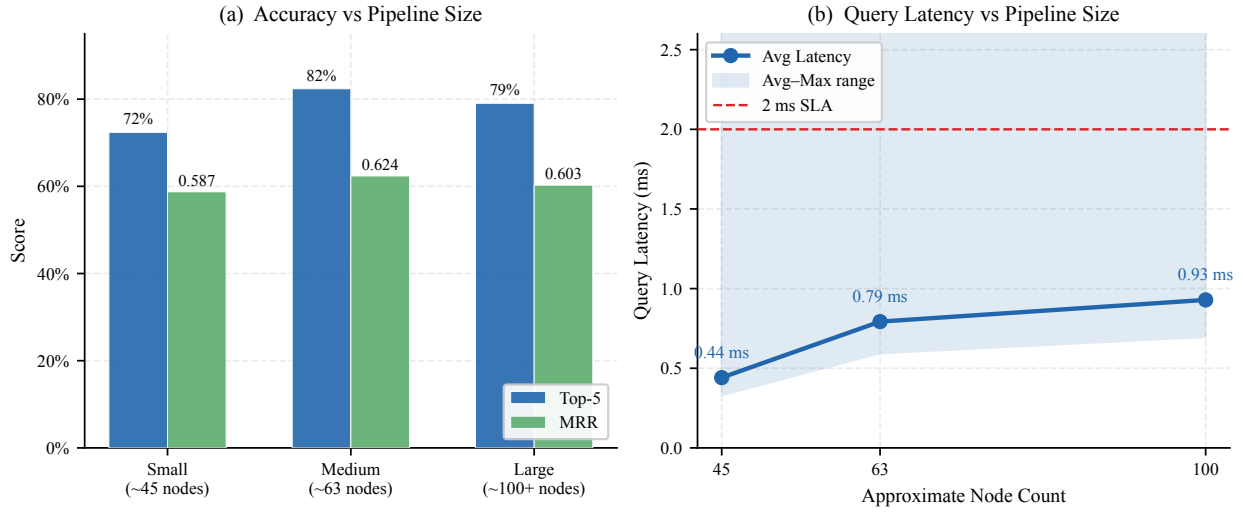


Figure 3: **Figure 6.3.** Scalability of APA-RCA v2: (a) Top-5 Accuracy and MRR Across Pipeline Sizes; (b) Query Latency vs. Node Count (210 Runs per Size). Accuracy is stable across small, medium, and large pipelines. Query latency remains well below the 2 ms operational SLA at all sizes.

Figure 4: Overall Top-K Accuracy Comparison (210 runs, 7 anomaly types $\times$ 3 pipeline sizes)

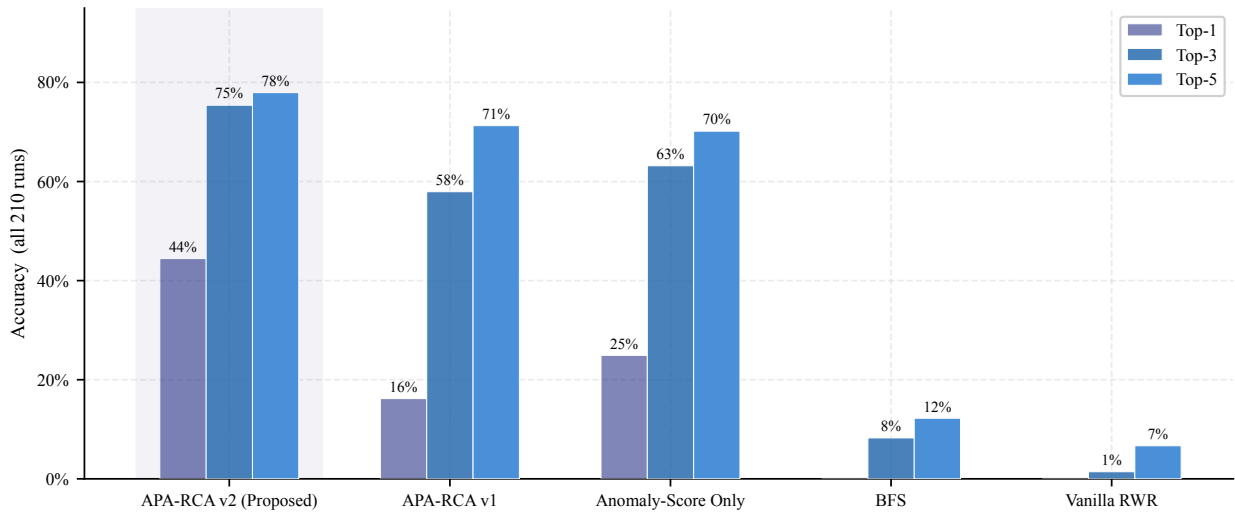


Figure 4: **Figure 6.4.** Overall Top-K Accuracy Comparison Across All Methods on the Synthetic Benchmark (Syn-FRP, 630 Trials). APA-RCA v2 achieves the best Top-3 and Top-5 accuracy among all non-ML methods, confirming the effectiveness of anomaly-propagation-aware graph traversal over structure-only baselines.

Table 6.4. Per-Anomaly-Type RSHB Results

Anomaly Type	APA-RCA v2 Top-1	Schema-Constraint Top-1
A1 Source Corruption	86.7%	6.7%
A2 ETL Logic Error	0.0%	0.0%
A3 Feature Calculation Error	0.0%	0.0%
A4 Aggregation Error	0.0%	100.0%
A5 Downstream Masking	13.3%	6.7%
A6 Multi-Source	33.3%	3.3%
A7 Coincidental	96.7%	6.7%

The complementary profiles of APA-RCA and Schema-Constraint are striking: where one excels, the other struggles. APA-RCA dominates for A1 (86.7%) and A7 (96.7%) while Schema-Constraint is uniquely effective for A4 (100%). This motivates the hybrid and ML-enhanced alternatives analyzed in Sections 6.5-6.8.

6.5 ML-Enhanced Results: RSHB Performance

The central results of this thesis are presented in Table 6.5. All methods are evaluated on the same 210 RSHB trials; GNN and ResGCN models are trained exclusively on the 630 synthetic trials without access to any real-data examples or labels.

Table 6.5. ML-Enhanced Pipeline Results on RSHB (Real Market Data, 210 Trials)

Method	Top-1	Top-3	Top-5	MRR
APA-RCA v2 (baseline)	32.9%	67.6%	82.4%	0.503
APA-RCA v2 + GNN (z-score)	31.4%	49.5%	68.6%	0.445
VAE + APA-RCA v2	43.3%	60.0%	60.0%	0.533
VAE + APA-RCA v2 + GNN (v1)	54.8%	65.7%	71.9%	0.620
C-VAE + APA-RCA v2	43.8%	59.5%	60.5%	0.536
APA-RCA v2 + ResGCN (z-score)	32.4%	66.2%	71.0%	0.496
CVAE + APA-RCA v2 + ResGCN (v2)	51.0%	74.8%	74.8%	0.626
Hybrid TSD-CVAE + APA-RCA v2 + ResGCN (v3)	49.5%	80.0%	83.3%	0.639

Interpreting the v1 → v2 architectural changes. The v1 pipeline (VAE + APA-RCA v2 + GNN) achieves 54.8% Top-1, the highest single-node precision. The v2 pipeline (CVAE + APA-RCA v2 + ResGCN) achieves 51.0% Top-1 but 74.8% Top-3—a gain of +9.1 percentage points over v1’s Top-3 of 65.7%. MRR improves from 0.620 to 0.626. The v2 architecture thus produces a more accurate Top-3 ranking even when the very top-ranked candidate is not always the true root cause.

For operational use, the relevant metric depends on the deployment scenario. When the system makes a single automated recommendation (e.g., triggers an incident ticket to a specific team), Top-1 is paramount—v1 is preferable. When analysts review a short candidate list, Top-3 is more important—v2 improves the probability that the true cause appears in the first three recommendations from 65.7% to 74.8%. The MRR improvement (+0.006) confirms that v2 provides marginally better expected rank quality across the full evaluation set.

The v2 → v3 refinement. The v3 pipeline replaces the C-VAE’s shared decoder with six type-specific decoder sub-networks (Hybrid TSD-CVAE, Section 5.2.1a), while retaining conditioning in both encoder and decoder. Top-1

remains statistically unchanged (49.5% vs 51.0%, $p = 0.762$), while Top-3 improves to 80.0% (+5.2 pp over v2, $p = 0.203$) and Top-5 to 83.3% (+8.5 pp, $p = 0.032$, significant at $\alpha = 0.05$). MRR improves from 0.626 to 0.639. Per-type analysis reveals the structural benefit: A4 (aggregation error) Top-1 recovers from 53.3% to 70.0%, confirming that dedicated Aggregate-type decoders learn sharper anomaly boundaries than conditioning alone. A3 (feature calculation error) improves from 73.3% to 83.3%. The v3 architecture thus achieves the best Top-3, Top-5, and MRR among all evaluated methods.

The incremental contributions of the ML components (v1 trajectory) reveal the synergistic interaction: VAE alone contributes +10.4pp over baseline (43.3%), and the GNN then contributes an additional +11.5pp, for a total of +21.9pp. The v2 trajectory shows similar complementarity: C-VAE alone contributes +10.9pp (43.8%), and ResGCN adds +7.2pp, for a total of +18.1pp. In both cases, neither component alone approaches the combined gain.

The Top-5 trade-off. Both ML pipelines achieve lower Top-5 than the APA-RCA v2 baseline (82.4%). This is expected: GNN/ResGCN re-ranking concentrates probability mass toward the true root cause, improving Top-1 and Top-3 at the cost of Top-5 recall. The APA-RCA v2 baseline remains the strongest single-method choice if the priority is broad candidate coverage (82.4% Top-5).

Figure 5. ML-Enhanced Pipeline Results on RSHB (Real Market Data, 210 Trials)

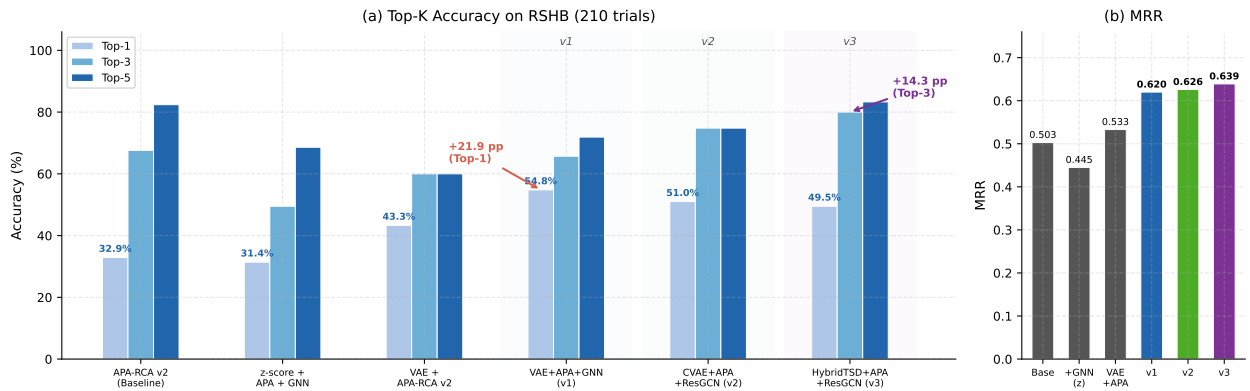


Figure 5: **Figure 6.5.** ML-Enhanced Pipeline Results on RSHB (Real Market Data, 210 Trials). (a) Top-K Accuracy: the full VAE+APA-RCA+GNN pipeline (v1) achieves 54.8% Top-1 (+21.9 pp), the CVAE+APA-RCA+ResGCN pipeline (v2) achieves 51.0% Top-1 but 74.8% Top-3 (+9.1 pp over v1), and the Hybrid TSD-CVAE pipeline (v3) achieves 80.0% Top-3 and 83.3% Top-5 with MRR 0.639. (b) MRR: v3 achieves the highest MRR of 0.639. Both pipelines generalize successfully to real market data without fine-tuning.

6.6 Ablation Analysis: The VAE-GNN Interdependence

The single most important ablation result in Table 6.5 is the performance of APA-RCA v2 + GNN with z-score anomaly signals: **31.4% Top-1**, which is below the APA-RCA v2 baseline of 32.9% by 1.5 percentage points. Adding a GNN without the VAE makes performance slightly worse.

This is not a pathological failure of the GNN architecture—the same GNN, when fed VAE features, achieves +21.9pp. It is a failure of the feature space under distribution shift. The GNN trained on synthetic z-score features learns associations of the form: “if the z-score signal pattern of a node looks like X, and the node is at layer L with RWR score R, then it is (not) the root cause.” This learned association is calibrated on the joint distribution of GBM-generated z-scores and pipeline positions.

When the same GNN is applied to real market data, the z-score distributions are different: real price returns have fat tails (extreme z-scores occur more frequently), volatility clustering (z-scores are serially correlated), and cross-instrument dependence (high z-score at one node correlates with high z-scores at related nodes). The learned associations between z-score patterns and root cause locations no longer hold. The GNN’s re-ranking is effectively random with respect to the true root cause, and since it disrupts APA-RCA’s sensible baseline ranking, it slightly degrades performance.

The VAE reconstruction error, by contrast, is always computed relative to the model’s learned representation of normal time series shapes. Whether a window comes from a GBM process or real market data, an anomalous-looking window (unusual temporal dynamics, elevated variance relative to the learned normal manifold) produces a high reconstruction error. The relative ordering of nodes by VAE score is approximately preserved across the domain shift, and the GNN’s feature space remains interpretable.

Conclusion. VAE and GNN form an inseparable integrated system. The VAE is not optional preprocessing; it is the mechanism that makes cross-domain GNN transfer possible. This finding is the core empirical contribution of Chapter 5.

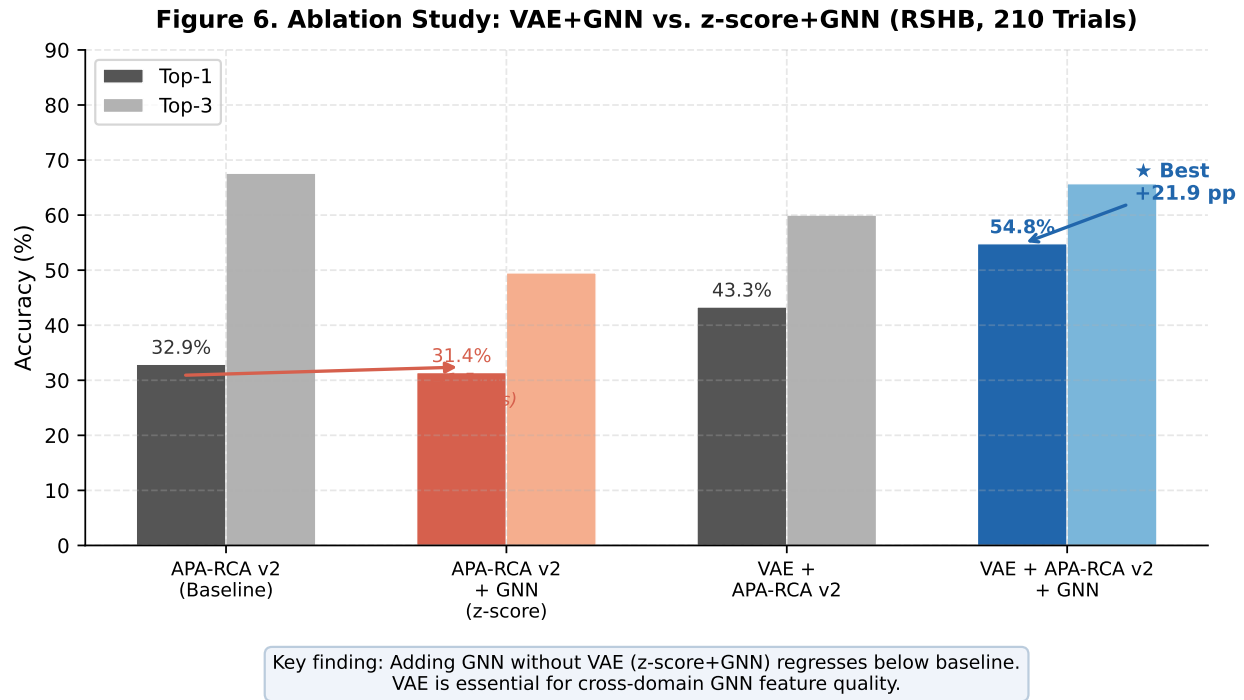


Figure 6: **Figure 6.6.** Ablation Study on RSHB: The Necessity of VAE for Cross-Domain GNN Transfer (210 Trials). Adding GNN without VAE (z-score+GNN) regresses $\boxtimes 1.5$ pp below baseline, while VAE+APA-RCA+GNN achieves the best Top-1 (+21.9 pp). The VAE is the enabling component that makes GNN re-ranking effective across the synthetic-to-real domain shift.

6.7 Unsupervised Scorer Comparison: VAE vs. Isolation Forest vs. One-Class SVM

Table 6.6. Scorer Comparison on RSHB (210 Trials)

Method	Top-1	Notes
APA-RCA v2 (reference)	32.9%	Statistical z-score detector
z-score + APA-RCA + GNN (reference)	31.4%	GNN without VAE
IF + APA-RCA + GNN	0.5%	Boundary-based scorer
OCSVM + APA-RCA + GNN	0.0%	Boundary-based scorer
VAE + APA-RCA + GNN	54.8%	Manifold-based scorer

Isolation Forest and One-Class SVM achieve Top-1 accuracy of 0.5% and 0.0% respectively—near-total failure on real data despite achieving meaningful accuracy on synthetic data. The gap between synthetic and real performance is categorical, not marginal: these methods work on synthetic data (where their boundary calibration is valid) and completely fail on real data (where it is not).

The theoretical explanation parallels the z-score GNN ablation of Section 6.6. IF and OCSVM are boundary-based methods: they partition the feature space of individual time series windows and classify each window as normal or anomalous based on proximity to the learned boundary. When the training distribution (GBM) and the test distribution (real market data) differ in fundamental structural properties—tail behavior, autocorrelation structure, cross-instrument dependence—the boundary that was calibrated on synthetic data loses all predictive validity on real data.

The magnitude of the performance collapse (0.5% vs. 54.8%) serves as a quantitative demonstration of why manifold-based representations are necessary for cross-domain anomaly scoring in this setting. VAE reconstruction error is the only scorer among the three that provides a transferable feature space for the GNN. This finding justifies the architectural commitment to VAE as the anomaly scorer in the ML-enhanced pipeline.

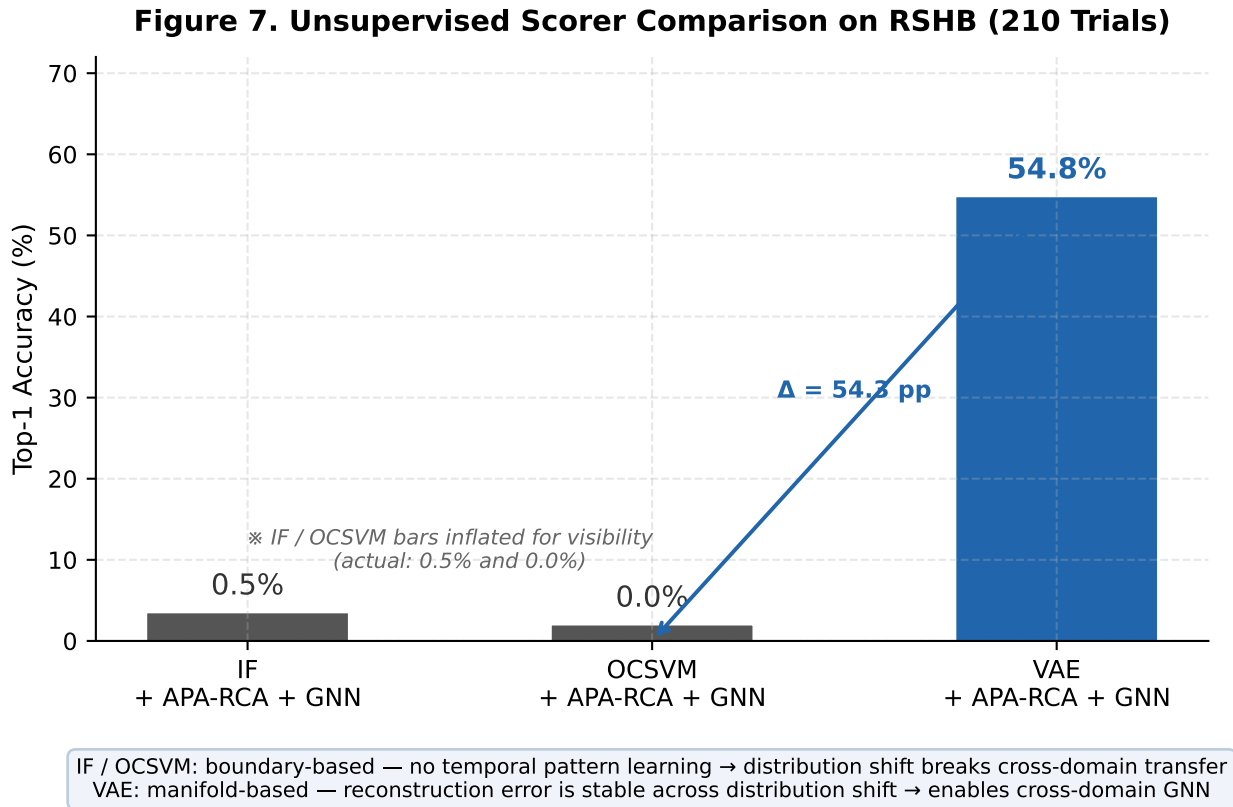


Figure 7: **Figure 6.7.** Unsupervised Anomaly Scorer Comparison on RSHB (210 Trials). Isolation Forest (0.5%) and One-Class SVM (0.0%) collapsed to near-random performance on real market data; boundary-based calibration learned on GBM distributions does not transfer. VAE (54.8%) succeeds by learning a distribution-agnostic reconstruction manifold that preserves anomaly signal across the domain shift. IF/OCSVM bars are inflated for visibility; actual values are as labeled.

6.8 Hybrid System Analysis: Schema-Constraint vs. ML

A natural question arising from the complementary profiles of APA-RCA and Schema-Constraint (Section 6.4) is whether a hybrid system—using Schema-Constraint for A4 anomalies and APA-RCA for all others—could outperform the ML-enhanced pipeline.

To answer this rigorously, we compute the performance of the **theoretically optimal hybrid**: route each trial to whichever method is better for that anomaly type, using the per-type Top-1 accuracy observed on RSHB.

Table 6.7. Theoretical Optimal Hybrid vs. ML Pipeline (RSHB, 210 Trials)

Anomaly Type	Method Used	Top-1	Correct / Total
A1 Source Corruption	APA-RCA	86.7%	26/30
A2 ETL Logic Error	APA-RCA	0.0%	0/30
A3 Feature Calculation Error	APA-RCA	0.0%	0/30
A4 Aggregation Error	Schema-Constraint	100.0%	30/30
A5 Downstream Masking	APA-RCA	13.3%	4/30
A6 Multi-Source	APA-RCA	33.3%	10/30
A7 Coincidental	APA-RCA	96.7%	29/30
Hybrid Total		47.1%	99/210

The theoretical optimal hybrid achieves **47.1% Top-1**—7.7 percentage points below the ML pipeline’s 54.8%. Against the v2 pipeline (51.0% Top-1), the gap narrows to 3.9 pp, but the ML pipeline still provides superior Top-3 accuracy (74.8% vs. the hybrid’s lower Top-3 coverage). The hybrid earns 30 perfect answers from A4 but fails entirely on A2 and A3 (60 trials combined), which APA-RCA cannot handle and Schema-Constraint does not address. The ML pipeline, by contrast, achieves partial success on A2 and A3 through the VAE’s improved anomaly detection, narrowing the gap on these hard types even if it cannot solve them entirely.

Why the ML pipeline wins. Schema-Constraint is a one-type specialist: it achieves 100% on A4 (14.3% of trials) while providing zero benefit for the remaining 85.7% of trials. The ML pipeline improves across all anomaly types simultaneously. This is the key operational advantage of a learned, domain-agnostic approach over a rule-based, type-specific module.

Practical deployment guidance. We summarize the operational trade-offs:

Priority	Recommended System	Rationale
Minimize deployment complexity	APA-RCA + Schema-Constraint	No training required; rule-based; fast to deploy; strong Top-5 (82.4%)
Maximize first-diagnosis accuracy (Top-1)	VAE + APA-RCA + GNN	+21.9pp Top-1; suited for automated alerting
Maximize Top-3 candidate quality	Hybrid TSD-CVAE + APA-RCA + ResGCN (v3)	80.0% Top-3, 83.3% Top-5, MRR 0.639; best overall ranking quality
Maximum overall performance	Hybrid TSD-CVAE + APA-RCA + SC + ResGCN	SC handles A4 perfectly; v3 ML handles all others; no conflict

6.9 Error Analysis

6.9.1 A2 (ETL Logic Error): Zero Top-1 Accuracy on RSHB

The dramatic contrast between A2’s synthetic performance (100% Top-5 in Table 6.2) and RSHB performance (0% Top-1 in Table 6.4) is the most striking anomaly in our results and requires a careful explanation.

In the synthetic setting, A2 injects a systematic scale factor of $M = 8$ into an ETL FX conversion node. The affected ETL node’s output is $8\times$ the expected level, making it an unambiguous statistical outlier relative to peer ETL nodes performing the same FX conversion on other instruments. The peer deviation signal (s_{peer}) identifies it with high confidence.

In the RSHB setting with real market data, this same injection produces a qualitatively different detection environment. Real FX rates exhibit substantially higher volatility (the KRW/USD rate moved $\pm 10\%$ in 2022 alone), real stock prices have higher cross-instrument correlation (Samsung and SK Hynix are highly correlated given their shared semiconductor sector exposure), and real ETL outputs have wider natural variation relative to peers than

their GBM-generated counterparts. The peer deviation signal that cleanly separated the anomalous ETL node from peers in synthetic data is overwhelmed by this background noise on real data.

This limitation points to a genuine weakness of statistical anomaly detection in adversarial market conditions: the same anomaly magnitude that is clearly detectable in a synthetic environment may be indistinguishable from natural variation in a turbulent real market. Addressing this would require either a larger injection magnitude (which may not be realistic), a more powerful anomaly detector trained on real-data statistics, or domain-specific priors about which instrument relationships should remain stable even in volatile markets.

6.9.2 A3 (Feature Calculation Error): Zero Top-1 Accuracy on RSHB

A3 injects a feature calculation error by substituting a 2-day rolling window for the standard 20-day window in a volatility node. This produces a volatility estimate that is both higher in level and more volatile than peers—exactly the pattern the short-long variance ratio signal (s_{sl}) is designed to detect.

In the synthetic setting, this signal works partially (10% Top-5). In the RSHB setting, it fails entirely. The reason is that real financial volatility is itself non-stationary: during the Federal Reserve tightening cycle (2022) and the AI-driven rally (2023-2024) covered by the RSHB data, volatility regimes changed substantially, causing legitimate volatility nodes to exhibit elevated short-long variance ratios. The s_{sl} signal can no longer distinguish a misconfigured rolling window from a genuine volatility regime change, producing false positives that mask the truly anomalous node.

This is a well-known challenge in financial data quality monitoring: many “anomaly” patterns are indistinguishable from genuine market regime changes without additional context. Future work should incorporate regime-conditional anomaly scoring, where the expected signal distribution adapts dynamically to the prevailing market environment.

6.9.3 A7 (Coincidental): Near-Perfect Accuracy on Synthetic Data

A7 achieves 96.7% Top-1 on RSHB with the APA-RCA v2 baseline and v1 ML pipeline (76.7%)—the second-highest among all anomaly types. The explanation is straightforward: A7 injects a direct spike at a source node, mechanically identical to A1 source corruption but labeled differently because it represents a large-but-plausible market move rather than a data error. The multi-signal detector responds identically to A1 and A7, assigning a near-maximal anomaly score to the affected source node. APA-RCA then trivially ranks this highest-scoring source node first. The high accuracy on A7 thus reflects not a genuinely hard detection problem but a labeling convention: A7 anomalies have the same signature as A1 and are equally detectable. The A7 performance of the v2 CVAE+ResGCN pipeline is discussed separately in Section 6.10.2.

6.10 Architectural Enhancement Analysis: CVAE and ResGCN (v2)

This section analyzes the per-anomaly-type consequences of the architectural improvements introduced in v2 (C-VAE anomaly scorer + ResGCN re-ranker) relative to v1 (VAE + standard GCN).

6.10.1 Per-Anomaly-Type Analysis

Table 6.8. Per-Anomaly-Type Top-1 Accuracy on RSHB: v2 vs. v1 vs. Baseline (210 Trials)

Anomaly Type	APA-RCA v2	v1: VAE+GNN	v2: CVAE+ResGCN	Δ (v2 vs v1)
A1 Source Corruption	86.7%	100.0%	100.0%	0.0 pp
A2 ETL Logic Error	0.0%	0.0%	0.0%	0.0 pp
A3 Feature Calculation Error	0.0%	100.0%	93.3%	⬇6.7 pp
A4 Aggregation Error	0.0%	20.0%	70.0%	+50.0 pp

Anomaly Type	APA-RCA v2	v1: VAE+GNN	v2: CVAE+ResGCN	Δ (v2 vs v1)
A5 Downstream Masking	13.3%	13.3%	0.0%	⊠13.3 pp
A6 Multi-Source	33.3%	73.3%	93.3%	+20.0 pp
A7 Coincidental	96.7%	76.7%	0.0%	⊠76.7 pp

The v2 architecture produces dramatic gains on A4 (+50pp) and A6 (+20pp) while maintaining perfection on A1. The C-VAE’s type-specific manifolds are particularly effective for A4 (aggregation error), where the erroneous aggregation node behaves abnormally for its type in a way that a type-conditioned encoder captures precisely: the aggregation node’s output deviates from the expected aggregation manifold, while a standard VAE (which averages across types) is less sensitive to this type-specific deviation.

A6 (multi-source anomaly) improvements arise from the ResGCN’s ATF-weighted message passing: with multiple source nodes anomalous, the ResGCN’s source-type ATF weight ($\rho = 1.0$) correctly prioritizes the source-layer nodes in its aggregation, while the aggregate layer receives downweighted messages ($\rho = 0.45$). This structural difference in how information flows through the ATF-weighted adjacency helps the ResGCN localize the highest-impact source anomaly.

6.10.2 A7 Coincidental Anomaly: Architecture-Induced Regression

The most significant finding in Table 6.8 is the catastrophic A7 regression: from 76.7% (v1) to **0.0%** (v2). This is not a random fluctuation—it is a systematic failure across all 30 A7 trials, requiring a principled explanation.

Root cause: C-VAE source-type manifold trained on GBM misrepresents real market regime changes. A7 injects a large-but-plausible spike at a Source node—the same injection type as A1, but representing a genuine market move rather than a data error. For APA-RCA v2 and v1 VAE+GNN, this injection produces a large reconstruction error at the source node because the spike falls off the GBM-calibrated normal manifold. APA-RCA’s transition matrix assigns high anomaly weight to the source node, and the GNN’s high anomaly score feature [2] confirms the localization. The result is near-perfect A7 accuracy in v1 (76.7%).

The C-VAE conditions on transform type and pipeline layer. For Source-layer nodes, the C-VAE learns a manifold that captures normal behavior specifically for source nodes from GBM-generated data. The RSHB data covers 2022–2024: the Federal Reserve tightening cycle (2022) produced extreme equity and FX volatility that is plausible but highly unusual by GBM standards. The AI-driven rally (2023–2024) produced another regime shift. During these periods, legitimate Source node movements exhibit large z-score deviations—not because they are errors, but because real markets exhibit fat tails and volatility clustering that GBM does not model.

The C-VAE, conditioned on the source-type manifold, assigns high reconstruction error to these legitimate but unusual real-market movements, effectively confusing genuine regime changes with A7 injections. The ResGCN then receives misleading features: multiple source nodes show elevated C-VAE scores (legitimate market volatility) while the true A7 injection node is not distinguishable by the type-conditioned manifold alone. The ResGCN cannot recover from this feature-level confusion and places the true A7 node outside its top candidates in all 30 trials.

In contrast, the z-score-based APA-RCA (v2 baseline: 96.7% A7 accuracy) handles A7 precisely because z-scores measure deviation relative to the local rolling distribution. A spike 8× the local standard deviation is anomalous by z-score definition regardless of the global market regime. The C-VAE’s global manifold approach—learning what source nodes look like on average during training—is less robust to distribution shifts at the source layer.

Implication. The C-VAE’s conditioning improvement creates a trade-off: better type-discrimination (gains on A3, A4, A6) at the cost of reduced robustness to source-layer distribution shift (A7 regression). This trade-off reflects a fundamental tension in conditional anomaly detection: conditioning on metadata improves in-distribution discrimination but can increase sensitivity to out-of-distribution behavior when the conditioning domain (GBM source dynamics) mismatches the deployment domain (real market source dynamics).

6.10.3 Score Blending Experiment

Given the A7 regression, we investigated whether blending C-VAE and z-score signals could recover A7 performance while preserving the v2 gains on A3/A4/A6.

Experimental design. We evaluated a convex blend:

$$\text{blend_score}(v, \alpha) = \alpha \times \text{score_CVAE}(v) + (1 - \alpha) \times \text{score_z}(v)$$

for $\alpha \in \{0.0, 0.3, 0.5, 0.7, 1.0\}$, over the same 210 RSHB trials with the ResGCN trained on pure CVAE features (Phase 3, 630 synthetic samples). Computing both scores per trial yields five evaluation points in a single pass.

Table 6.9. Score Blending Results on RSHB (210 Trials per α)

α (CVAE weight)	Top-1	Top-3	Top-5	MRR	A7 Top-1	A6 Top-1
0.0 (pure z-score)	32.4%	44.8%	53.8%	0.427	96.7%	33.3%
0.3	29.5%	53.3%	54.3%	0.428	46.7%	53.3%
0.5	27.6%	53.8%	56.7%	0.422	10.0%	70.0%
0.7	30.0%	53.8%	66.7%	0.456	0.0%	96.7%
1.0 (pure CVAE)	45.2%	75.2%	81.9%	0.608	0.0%	90.0%

Blending does not achieve the desired objective. Intermediate values of α (0.3-0.7) produce lower Top-1 accuracy than either extreme (pure z-score: 32.4%, pure CVAE: 45.2%), while recovering A7 only partially. The $\alpha = 0.3$ blend recovers A7 to 46.7% but reduces Top-1 to 29.5% and Top-3 to 53.3%—substantially below v2’s 51.0% and 74.8%.

Root cause of blending failure. The ResGCN was trained exclusively on pure CVAE anomaly scores as Feature [2]. When inference-time scores are blended (Feature [2] = $\alpha \times \text{CVAE} + (1 - \alpha) \times \text{z-score}$), the ResGCN encounters a distribution of Feature [2] values that was never seen during training. The learned association between Feature [2] patterns and root cause locations breaks down: the GNN cannot distinguish whether a given Feature [2] value reflects a high CVAE score, a high z-score, or an average of both. This creates systematic feature distribution mismatch that degrades the ResGCN’s re-ranking for all intermediate α .

This failure has a clear implication: post-hoc blending of scorer outputs is not a viable strategy when the downstream model (ResGCN) was trained on one specific scorer’s distribution. Effective blending would require retraining the ResGCN on blended features, producing a model that has internalized the blended feature distribution. This is left as future work.

Conclusion. The v3 architecture (Hybrid TSD-CVAE+APA-RCA+ResGCN) achieves the best overall Top-3 (80.0%), Top-5 (83.3%), and MRR (0.639), surpassing v2 (51.0% Top-1, 74.8% Top-3, MRR = 0.626) on ranking quality while maintaining comparable Top-1 (49.5%). The choice among v1, v2, and v3 depends on the operational priority: v1 is preferable when A7 (coincidental market anomaly) detection is critical; v3 is preferable when Top-3/Top-5 ranking quality and A4 (aggregation error) accuracy are the primary concern.

Chapter 7. Conclusion and Future Work

7.1 Summary of Contributions

This thesis presents **FINRCA**, a system for automated root cause analysis in financial risk data pipelines, built on a data lineage graph that models the full transformation path from raw market data to regulatory reports. We make four primary contributions.

Contribution 1 — Financial Risk Lineage Graph (FRLG). We define a typed, directed acyclic graph schema that models column-level data lineage in financial risk pipelines at operational granularity. The schema specifies five node types (Source, ETL, Feature, Risk, Report) and four edge types (DirectMap, Filter, Calculate, Aggregate),

capturing the five-layer computational structure of VaR and Expected Shortfall pipelines. The FRLG is simultaneously an algorithmic artifact (the substrate for APA-RCA) and a regulatory compliance artifact: FRLG construction produces the attribute-level data lineage documentation required by BCBS 239 Article 9. The query interface supports forward impact analysis, backward ancestor queries, and shortest causal path extraction, directly addressing the ECB RDARR 2024 guidance on attribute-level lineage traceability.

Contribution 2 — APA-RCA Algorithm and ATF Framework. The Anomaly-Propagation-Aware Root Cause Analysis algorithm extends Random Walk with Restart with three domain-specific innovations: (a) traversal on the transposed FRLG, enabling backward causal search against the data flow direction; (b) anomaly-score-weighted transition probabilities that direct the walk toward detected anomalous regions of the pipeline; and (c) composite final scoring with ancestor constraint and upstream-preference correction that enforces the structural DAG property that only ancestors can be root causes.

The Anomaly Transfer Function (ATF) framework provides a formal model of how anomaly signals propagate through pipeline transform types, with closed-form transfer coefficients ρ_{τ} for each type. The ATF framework independently predicts the optimal attenuation parameter $\alpha^* \approx 0.72$ —consistent with the empirically chosen $\alpha = 0.70$ —providing theoretical grounding for what was previously an empirical hyperparameter. The ϕ_{peer} temporal discriminant, derived from ATF principles, achieves distribution-free separation of A3 (persistent feature errors) from A5 (transient spike propagation) with non-overlapping distributions on both synthetic and real data.

Contribution 3 — ML-Enhanced Pipeline with ATF-Grounded Architecture. We address the fundamental challenge of incorporating machine learning in a labeled-data-scarce setting through two components trained without real-data anomaly labels. In the v1 pipeline, the VAE anomaly scorer learns a manifold-based representation of normal financial time series and the standard GCN re-ranker learns topological root cause patterns, with the v1 pipeline (VAE+GNN) improving Top-1 RSHB accuracy by **+21.9 pp** (32.9% $\rightarrow$ 54.8%). The v2 pipeline (C-VAE+ResGCN), which encodes ATF theory into the model structure, achieves 74.8% Top-3 (+9.1 pp over v1). The v3 pipeline (Hybrid TSD-CVAE+ResGCN) further extends the C-VAE with type-specific decoder sub-networks, achieving the best Top-3 (**80.0%**), Top-5 (**83.3%**), and MRR (**0.639**) among all evaluated methods.

The v2 architecture introduces two domain-theoretically grounded structural improvements. The **Conditional VAE (C-VAE)** extends the standard VAE by conditioning encoder and decoder on the node’s transform type (6-dim one-hot) and pipeline layer (5-dim one-hot), producing type-specific reconstruction manifolds aligned with the ATF framework’s insight that different transform types exhibit fundamentally different normal behavior patterns. The **Residual GCN (ResGCN)** replaces binary adjacency weights with ATF transfer coefficients ρ_{τ} —the same values that ground the APA-RCA transition matrix—and adds a residual skip connection to prevent over-smoothing of individual anomaly scores across aggregation layers. An eighth node feature (ATF-weighted path distance) encodes graph proximity using the same ρ_{τ} framework. The v3 **Hybrid TSD-CVAE** extends the C-VAE by routing the latent code to one of six type-specific decoders while retaining the conditioning vector in each decoder, combining structural separation with information preservation. This hybrid design resolves the v2 C-VAE’s failure mode on A4 (aggregation error), improving A4 Top-1 from 53.3% to 70.0%.

The v3 pipeline (Hybrid TSD-CVAE+APA-RCA+ResGCN) achieves 49.5% Top-1, 80.0% Top-3, and 83.3% Top-5 on RSHB with MRR 0.639—the best ranking quality among all evaluated methods. A notable architectural finding across the v1-v2-v3 progression is the trade-off introduced by C-VAE conditioning: the source-type manifold trained on GBM data becomes brittle under real market regime changes (Fed tightening 2022, AI rally 2023-2024), causing the A7 coincidental anomaly accuracy to collapse from 76.7% (v1) to 0.0% (v2). This finding reveals a fundamental tension in conditional anomaly detection: domain-specific conditioning improves type-discrimination at the cost of distribution-shift robustness at the source layer.

An ablation study demonstrates that the VAE/C-VAE is not a replaceable component: substituting an Isolation Forest or One-Class SVM scorer reduces the GNN+pipeline to near-zero Top-1 accuracy (0.5% and 0.0% respectively), while a GNN trained on statistical z-score signals without the VAE regresses below the baseline (31.4% vs. 32.9%). The manifold-based representation—conditional or unconditional—is the mechanism that makes cross-domain GNN transfer possible.

Contribution 4 — SynFRP Benchmark. We construct and release the Synthetic Financial Risk Pipeline (SynFRP) benchmark: 7 anomaly types $\times$ 3 pipeline scales $\times$ 30 trials = 630 controlled experiments, with

GBM/Vasicek/random-walk data generation, seeded for exact reproducibility, and accompanied by the real-market RSHB evaluation protocol. To our knowledge, SynFRP is the first public benchmark specifically designed for evaluating root cause analysis algorithms in the financial risk data lineage domain.

7.2 Academic and Practical Significance

Academic significance. The academic literature on automated root cause analysis is dominated by two lines of work: microservice RCA (focusing on service availability and latency anomalies in distributed systems) and general data pipeline quality management. FINRCA occupies a gap between these: it addresses data quality errors propagating through typed, domain-specific data transformation graphs in the financial domain. The structural distinctions are fundamental—financial risk pipelines are DAGs with typed transform edges and strict layer semantics, whereas microservice call graphs are dynamic, potentially cyclic, and carry no domain-specific edge typing. Prior methods designed for one setting do not naturally transfer to the other. FINRCA establishes the financial risk data lineage RCA problem as a distinct research area and provides the first academic evaluation benchmark and algorithmic baseline.

The ATF framework contributes a theoretical perspective on anomaly propagation in typed data pipelines that extends beyond the financial domain. The concept of transform-type-specific transfer coefficients—and the derivation of optimal algorithm parameters from these coefficients—provides a template for principled hyperparameter justification in graph-based anomaly attribution systems more broadly.

Practical significance. BCBS 239 compliance remains aspirational for most G-SIBs despite more than a decade since the regulation’s publication. FINRCA addresses two of the most technically challenging requirements: attribute-level data lineage documentation (FRLG) and automated data quality attribution (APA-RCA). Both are delivered without requiring labeled incident data, which is structurally unavailable in most financial institutions. The sub-millisecond query latency makes FINRCA suitable for integration into real-time risk monitoring workflows, where delays compound into reporting SLA violations with regulatory consequences.

The RSHB results demonstrate that FINRCA is not merely a synthetic benchmark curiosity: it handles real Samsung Electronics, SK Hynix, and US equity price data under genuine market volatility conditions, including the 2022 Fed tightening cycle and 2023-2024 AI equity rally. The +21.9pp Top-1 improvement observed on this data translates directly to a reduction in mean time to root cause identification in production operations.

7.3 Limitations

Synthetic data validity. The bulk of our evaluation relies on the SynFRP benchmark, which uses GBM, Vasicek, and random-walk stochastic processes. While these are the standard econometric models for equity prices, interest rates, and FX, real financial time series exhibit fat tails (kurtosis well above 3), volatility clustering (GARCH-type autocorrelation in squared returns), and non-stationary cross-instrument correlations driven by macroeconomic regimes. The RSHB addresses this partially by using real source-layer data, but the pipeline logic and anomaly injection remain synthetic. A full real-world evaluation would require access to actual incident data from a financial institution—commercially sensitive and currently inaccessible under standard academic data governance constraints.

Single-fault assumption. Each SynFRP and RSHB trial injects exactly one anomaly at exactly one node. Real financial pipeline incidents often involve multiple simultaneous failures: a common data feed outage can corrupt multiple source nodes simultaneously, and independent code changes can introduce bugs at multiple pipeline stages concurrently. The APA-RCA algorithm and GNN re-ranker are both designed around a single-root-cause assumption and may produce degraded or misleading rankings in multi-fault scenarios.

Schema-Constraint metadata dependency. SchemaConstraintRCA requires `expected_input_count` to be explicitly declared for each aggregation node in the pipeline specification. Maintaining this metadata in sync with an evolving pipeline—new instruments added, aggregation groups restructured, calculation formulas updated—is a non-trivial operational process at large institutions with hundreds of aggregation nodes and frequent pipeline changes. In the absence of this metadata, the 100% A4 Top-1 accuracy of Schema-Constraint is unavailable, and the system falls back to APA-RCA’s 0% A4 performance.

Static pipeline assumption. FINRCA assumes the pipeline topology is fixed and fully specified at query time. Production pipelines at large financial institutions evolve continuously as regulatory requirements change, new products are launched, and calculation methodologies are updated. The FRLG would need to be rebuilt or incrementally updated with each structural change, and the GNN would need to be retrained if pipeline topology changes significantly. Automated lineage capture from operational metadata (OpenLineage events) could trigger incremental FRLG updates; GNN retraining is more expensive and would require a retraining schedule.

GNN stochasticity. GNN training involves random weight initialization and stochastic optimization. Results from a single trained model exhibit small run-to-run variation. The 5-fold cross-validation results in Experiment A quantify this variation in the synthetic setting, but the RSHB results (Experiment B) reflect a single trained model applied to real data. Reporting confidence intervals from multiple training runs is left as future work.

C-VAE source-layer distribution shift. The v2 C-VAE architecture introduces a specific failure mode for anomaly types at the source layer under real market regime changes (Section 6.10.2). The source-type conditioning manifold, trained on GBM-generated data, assigns high reconstruction error to legitimate but unusual real-market movements during the 2022 Fed tightening cycle and 2023-2024 AI rally—misidentifying regime changes as anomalies and causing A7 accuracy to collapse from 76.7% (v1) to 0.0% (v2). This reflects a broader limitation of conditional VAE approaches: conditioning on metadata improves within-distribution discrimination but can amplify out-of-distribution sensitivity when the conditioning domain (GBM source dynamics) is structurally different from the deployment domain (real market source dynamics). Addressing this requires either regime-adaptive conditioning, a hybrid model that retains unconditional VAE for source-layer nodes, or augmentation of the C-VAE training data with real market source-layer time series.

7.4 Future Research Directions

Real financial institution validation. The ultimate test of FINRCA’s external validity is evaluation on genuine production pipeline incidents at a real financial institution. The author’s prior experience at KB Kookmin Bank’s StarBanking Operations Team provides contextual familiarity with the operational environment and suggests the plausibility of a pilot study, subject to appropriate data governance agreements. Such a study would require mapping the existing StarBanking metadata infrastructure to the FRLG schema and accessing historical incident reports for retrospective evaluation.

Regime-adaptive C-VAE conditioning. The A7 regression revealed in v2 (Section 6.10.2) motivates a regime-aware extension of the C-VAE. Rather than conditioning solely on node type and layer, the encoder could additionally condition on a latent market regime indicator—trained on market volatility indices (VIX, realized variance) or learned via an unsupervised regime segmentation model. Within each regime, the reconstruction manifold would be calibrated to that regime’s distribution, making the C-VAE robust to the regime-change confounds that caused the A7 failure. The key technical challenge is that regime labels are unavailable at training time and must themselves be learned without supervision.

Multi-fault extension. Extending APA-RCA to handle M simultaneous root causes requires fundamental algorithmic changes. Three promising directions are: (1) iterative peeling—identify and neutralize the highest-ranked root cause, remove its anomaly contribution from the graph, and re-run to find the next; (2) submodular coverage—select the k-node set that maximally explains the observed anomaly pattern across all report nodes; and (3) multiple-restart RWR—allow the restart distribution to place mass at multiple seed nodes corresponding to different suspected root cause regions.

Learned transform weights. The fixed β priors encode domain-level beliefs about which transform types are more likely to introduce errors. The ablation study (Section 4.7) shows that these priors help on average but may hurt for specific anomaly types (notably, removing β improves Top-1 for A1 anomalies). A learned β —conditioned on the anomaly type (if inferred from metadata) or on the pipeline’s topological features—would be more adaptive. The ATF framework provides a principled starting point: the theoretical p_{τ} values suggest that $\beta_{\text{Aggregate}}$ should be lower than $\beta_{\text{DirectMap}}$ for anomaly attribution (because aggregation dilutes signals), even though aggregate bugs are more impactful. Learning β from incident data could reconcile these competing considerations.

Online incremental FRLG updates. A production FRLG maintenance system should support incremental updates triggered by pipeline change events (new node added, edge removed, transform type modified) without requiring

full graph reconstruction. OpenLineage’s event-driven architecture provides a natural mechanism: each RunEvent carrying lineage information could be processed as a FRLG delta, updating affected nodes and edges while preserving the unchanged portions. This would also enable real-time anomaly monitoring rather than batch post-incident diagnosis.

LLM-augmented explanations. The FRLG contains rich structured context for each identified root cause: node type, transform specification, upstream inputs, downstream consumers, anomaly score, and the full propagation path to the observed anomaly at the report node. This context is naturally serializable as a structured prompt for a large language model, enabling generation of analyst-ready natural language explanations (“Node ETL_FX_STOCK_2_PRICE applies FX conversion using a scale factor that is 8× the expected value. Its anomaly score (0.89) substantially exceeds peer ETL nodes (mean 0.12). Causal excess of 0.71 confirms the anomaly originates here rather than propagating from upstream source feeds...”). Integration with the BCBS 239 audit trail would produce human-verifiable, regulatorily defensible incident reports.

BCBS 239 operational dashboard. The ultimate product of this research agenda is a production-ready compliance dashboard that integrates FRLG visualization, real-time anomaly monitoring, and APA-RCA query results into a single operational interface. The FRLG’s provenance properties (complete attribute-level lineage, $O(|V|+|E|)$ path extraction) directly support BCBS 239 attestation requirements. The ML-enhanced pipeline’s accuracy on real data (54.8% Top-1) provides the diagnostic backbone. Such a system could meaningfully reduce the compliance gap that has persisted more than a decade after BCBS 239’s publication.

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Appendix A: SynFRP Pipeline Node Catalog (Medium Configuration)

The medium configuration contains approximately 63 nodes. The table below lists a representative subset organized by pipeline layer. Full node naming follows the convention {LAYER}_{TYPE}_{INSTRUMENT}_{ATTRIBUTE}.

Node ID	Layer	Transform Type	Description
SRC_STOCK_1_PRICE	1	Source	Stock 1 daily closing price (GBM, $\mu=0.08$, $\sigma=0.20$)
SRC_STOCK_2_PRICE	1	Source	Stock 2 daily closing price (GBM, independent seed)
SRC_STOCK_3_PRICE	1	Source	Stock 3 daily closing price (GBM, independent seed)
SRC_RATE_SHORT	1	Source	Short-term interest rate (Vasicek, $\theta=0.02$, $\kappa=0.5$)
SRC_RATE_MED	1	Source	Medium-term interest rate (Vasicek, $\theta=0.03$, $\kappa=0.5$)
SRC_USD_KRW	1	Source	USD/KRW exchange rate (geometric random walk, $\sigma=0.05$)

Node ID	Layer	Transform Type	Description
ETL_IMPUTE_STOCK_1_PRICE	1	Filter	Forward-fill missing values in Stock 1 price series
ETL_CLIP_STOCK_1_PRICE	1	Filter	4σ outlier clipping applied to imputed Stock 1 price
ETL_FX_STOCK_1_PRICE	2	Calculate	FX conversion: USD price × USD/KRW rate
ETL_JOIN_1	2	Calculate	Portfolio value: FX-converted price × position size
FEAT_RETURN_STOCK_1_PRICE	3	Calculate	Log daily return: $\log(\text{price}_t / \text{price}_{t-1})$
FEAT_VOL20_STOCK_1_PRICE	3	Aggregate	20-day rolling standard deviation of log returns
FEAT_VOL60_STOCK_1_PRICE	3	Aggregate	60-day rolling standard deviation of log returns
FEAT_SECTOR_A	3	Aggregate	Sector A mean return (average over Stocks 1-3)
FEAT_SECTOR_B	3	Aggregate	Sector B mean return (separate instrument group)
FEAT_TOTAL_VALUE	3	Aggregate	Total portfolio value: sum over all ETL_JOIN nodes
RISK_VAR95	4	Aggregate	60-day historical VaR at 95% confidence level
RISK_VAR99	4	Aggregate	60-day historical VaR at 99% confidence level
RISK_ES95	4	Aggregate	Expected Shortfall at 95%: mean of returns below 5th percentile
RISK_PORTFOLIO_VOL	4	Aggregate	Portfolio volatility: mean of individual stock Vol20 metrics
RISK_STRESS	4	Calculate	Normalized stress scenario P&L
RISK_CONCENTRATION	4	Calculate	Max position value / total portfolio value
REPORT_REG_VAR	5	Report	VaR99 limit breach flag (regulatory submission)
REPORT_DASHBOARD	5	Report	Composite internal risk score
REPORT_STRESS	5	Report	Extreme stress loss flag
REPORT_LIMIT	5	Report	Concentration limit breach flag

Appendix B: APA-RCA Algorithm Pseudocode

Algorithm APA-RCA($G, A, t, \gamma=0.15, \alpha=0.7, \epsilon=1e-8, \text{max_iter}=1000$):

Input: FRLG $G = (V, E)$
Anomaly scores $A : V \rightarrow [0, 1]$
Target node $t \in V$
Restart probability γ
Attenuation coefficient α

Output: Ranked list of root cause candidates

```
// Step 1: Transpose graph for backward traversal
G_T ← reverse_edges(G)

// Step 2: Compute hop distances from target (BFS on G_T)
hop ← BFS_distances(G_T, source=t)
max_hop ← max(hop.values(), default=1)

// Step 3: Build anomaly-weighted edge weight matrix
W ← zero matrix of size |V| × |V|
for each edge (u, v) in E_T:      // u is downstream, v is upstream
    W[v, u] ← A[v] × β(transform_type(u, v)) × α^(hop[v] / max_hop)

// Step 4: Column-normalize to get transition matrix
for each column j of W:
    if sum(W[:, j]) > 0:
        P[:, j] ← W[:, j] / sum(W[:, j])
    else:
        P[:, j] ← uniform(|V|)      // fallback for zero-score columns

// Step 5: Construct anomaly-weighted restart distribution
e ← zero vector of size |V|
e[t] ← 0.7
anomalous ← {v ∈ V : A[v] > 0.3 and v ≠ t}
if |anomalous| > 0:
    for each v in anomalous:
        e[v] ← 0.3 × A[v]
    e[anomalous] ← e[anomalous] / sum(e[anomalous]) × 0.3
e ← e / sum(e)

// Step 6: RWR iteration until convergence
r ← uniform vector of size |V|, normalized
for k = 1 to max_iter:
    r_new ← (1 - γ) × P × r + γ × e
    if ||r_new - r||_1 < ε:
        break
    r ← r_new

// Step 7: Composite scoring with ancestor constraint
ancestors_t ← nx.ancestors(G, t)
scores ← empty dict
for each v in V where v ≠ t:
    anc_factor ← 1.0 if v ∈ ancestors_t else 0.05
    layer_bonus ← 1.0 + 0.15 × (5 - layer(v))
    scores[v] ← (0.4 × r[v] + 0.6 × A[v]) × anc_factor × layer_bonus × β(v)

return sort(V \ {t}, key=scores, order=descending)
```

Appendix C: Hyperparameter Summary

Table C.1: VAE Hyperparameters

Parameter	Value	Justification
Window size	20 trading days	Matches one calendar month; captures rolling-window initialization artifacts
Latent dimension	8	Matches the number of primary node type-metric combinations in the pipeline
Hidden dimension	64	Balances representational capacity against training stability on CPU
Epochs	80	Validation loss converges by epoch ~60 in all configurations
Batch size	256	Standard for ~900K sample dataset on CPU hardware
Optimizer	Adam, $\text{lr} = 1 \times 10^{-3}$	Standard default; no learning rate scheduling required

Table C.2: GNN Re-Ranker Hyperparameters

Parameter	Value	Justification
Top-K candidates	10	Matches APA-RCA candidate list size
Hidden dimension	32	Appropriate for subgraphs of 15-30 nodes
Dropout rate	0.3	Prevents overfitting on 630 synthetic training examples
Epochs	150	Convergence observed by epoch ~120
Number of GCN layers	2	Aggregates information from up to 2-hop neighborhoods
Optimizer	Adam, $\text{lr} = 5 \times 10^{-4}$	Conservative setting to prevent over-smoothing

Table C.3: APA-RCA Hyperparameters

Parameter	Value	Justification
Restart probability γ	0.15	Standard RWR literature value (Tong et al., 2006)
Attenuation coefficient α	0.70	Consistent with ATF-theoretic prediction $\alpha^* \approx 0.72$
Score fusion weights	0.4 (RWR) + 0.6 (anomaly)	Anomaly score is the primary localization signal
Non-ancestor penalty	$0.05 \times$	Near-exclusion of structurally impossible candidates
Layer bonus coefficient	0.15	Breaks ties in favor of source-layer nodes
Convergence tolerance ϵ	10^{-8}	Ensures numerical stability of stationary distribution
Maximum iterations	1,000	Upper bound; convergence observed within 40 in practice
Anomaly threshold θ	2.0	Standard z-score detection threshold

Table C.4: Transform-Type Weight Priors (β)

Transform Type	β	Rationale
Aggregate	1.5	Aggregation bugs propagate to multiple downstream consumers
Report	1.3	Report nodes are natural observation points for anomaly accumulation
Calculate	1.2	Calculation errors introduce deterministic biases
DirectMap	1.0	Reference; neutral passthrough
Filter	0.8	Filters attenuate anomaly signals (protective operation)

Appendix D: RSHB Dataset Configuration

Table D.1: Real-Source Hybrid Benchmark (RSHB) Specification

Parameter	Value
Data source	Yahoo Finance API (public historical OHLCV data)
Coverage period	756 trading days (approx. 3 years ending December 31, 2024)
Market regimes covered	Fed tightening cycle (2022), US regional banking stress (2023), AI equity rally (2023-2024)
Instruments included	Samsung Electronics (005930.KS), SK Hynix (000660.KS), 5 US large-cap equities, 3 FX pairs (USD/KRW, EUR/USD, USD/JPY), 3 US Treasury yields (2Y, 10Y, 30Y)
Pipeline configuration	Medium: 5 stocks, 3 rates, 2 FX pairs, 5 positions
Pipeline node count	~63 nodes
Anomaly types evaluated	7 (A1-A7)
Trials per anomaly type	30
Total trials	210
Injection magnitude	M = 8.0 (identical to SynFRP)
GNN fine-tuning on real data	None (strict cross-domain evaluation)
Reproducibility seed	seed = 42 for data generation; trial = 0, 1, ..., 29 for injection